

Production performance and biochemical profile of sea lettuce (*Ulva lactuca*): Influence of site-specific ecological factors and cultivation strategies

Farjana Akter Chamily^a, Md Mohiuddin^a, Mohammad Shakline Mostakim^a, Md Moshir Rahman^b, Sazeed Mehrab Souhardya^c, Md Nahiduzzaman^c, Md Sadequr Rahman Khan^a, Md Asaduzzaman^{a,*}

^a Department of Marine Bioresource Science, Faculty of Fisheries, Chattogram Veterinary and Animal Sciences University, Khulshi 4225, Chattogram, Bangladesh

^b Fish Conservation and Culture Laboratory, Department of Biological and Agricultural Engineering, University of California Davis, 17501 Byron Hwy, Byron, CA 94514, USA

^c ECOFISH-II Activity, WorldFish, Bangladesh and South Asia Office, Gulshan-2, Dhaka, Bangladesh

ARTICLE INFO

Keywords:

Macroalgae, *Ulva lactuca*
Cultivation systems
Mariculture
Nutritional compositions
Blue economy

ABSTRACT

Production performance and biochemical characteristics of seaweed under varying cultivation strategies and ecological conditions provide crucial insights for improving sustainable farming practices. Therefore, this study aims to investigate the production performance and biochemical profile of sea lettuce (*Ulva lactuca*) under three farming systems, namely floating long-line (FLL), off-bottom long-line (OBLL), and off-bottom net (OBN), and three site-specific (Khurushkul, Nuniachora, and Sonapara) ecological conditions for a period of nine months (October 2022 to June 2023). Depending on the site suitability for different cultivation systems, 15 plots (5 FLL, 5 OBLL, and 5 OBN) were established in Nuniachora, 10 plots (5 FLL, 5 OBLL) were established in Sonapara, and 5 plots (FLL) were established in Khurushkul of the Cox's Bazar district, Bangladesh. Depending on the cultivation months, we partially harvested *U. lactuca* every 15–30 days once it reached 25–35 cm in length. Results showed that the gross weight biomass (GWB) of *U. lactuca* cultured in the FLL ($F = 200.43, p < 0.001$) cultivation system was 45.38% and 39.85% higher compared to OBN and OBLL, respectively. The average daily growth rate (DGR) was found to be 11%–13% higher in the FLL method ($F = 236.11, p < 0.001$). Location-based comparisons indicated a significantly higher ($p < 0.05$) *U. lactuca* production in Khurushkul, followed by the Nuniachora, and the lowest was observed in Sonapara. Seasonality influenced the production performances of *U. lactuca*, with significantly higher ($p < 0.05$) growth observed from December to March due to favourable physio-chemical parameters. Multivariate analysis revealed that high salinity, pH, and nutrient concentrations positively influenced growth performances, while high temperature, turbidity, and total suspended solids (TSS) had negative effects. *U. lactuca* harvested from the FLL and OBN systems contained 14.3% and 9.44% higher amounts of protein ($F = 36.77, p < 0.001$) than the OBN system. The fatty acid analysis showed that *U. lactuca* from the FLL and OBN systems contained a significantly lower amount of saturated fatty acids but a higher amount of monounsaturated and n-6 polyunsaturated fatty acids compared to the OBLL system. Among the heavy metals and minerals analyzed, *U. lactuca* harvested from the FLL system contained significantly ($p < 0.05$) higher amounts of Na and K and lower amounts of Pb, Cr, Co, and Fe. The findings of the present study underscore the superiority of the floating long-line cultivation system for *U. lactuca*, highlighting its potential for sustainable farming practices to address global food security and utilization in various industrial applications.

1. Introduction

According to the United Nations, the world population is growing

rapidly and is expected to reach 9.7 billion by 2050 (De Wrachien et al., 2021; Sadigov, 2022). This population rise, together with changing dietary trends and increased urbanization, presents enormous challenges

* Corresponding author.

E-mail address: a.zamanbau@yahoo.com (M. Asaduzzaman).

<https://doi.org/10.1016/j.aquaculture.2024.741436>

Received 22 April 2024; Received in revised form 5 July 2024; Accepted 31 July 2024

Available online 3 August 2024

0044-8486/© 2024 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

to global food security and environmental sustainability (Fróna et al., 2019). In this context, there is a rising realization of the importance of diversifying food sources and developing sustainable agriculture techniques that may fulfil the nutritional demands of a growing population while minimizing environmental impact. However, the availability of land-based resources is constrained to fulfil the increasing demand for food, and the recognition of alternative, sustainable resources has become increasingly important (Bruhn et al., 2011). Cultivating seaweed stands as a viable alternative to traditional land-based agricultural practices for sustainable food supply without harming land resources, as it has rapid growth, high biomass, and cost-effective production prospects (Morais et al., 2020). Worldwide, there are around 2200 species of green seaweed, 6100 species of red seaweed, and 1800 species of brown seaweed (Morais et al., 2020). Among these seaweed species, around 250 species are utilized worldwide for commercial purposes, including food and phycocolloid manufacture (Chowdhury et al., 2022). According to the Food and Agriculture Organization, the global output of seaweed is anticipated to have surpassed 35 million tonnes, exceeding US\$ 16.5 billion annually (FAO, 2022). Most of this production, around 97%, originated from aquaculture, mostly marine aquaculture (FAO, 2022). The global area suitable for seaweed farming encompasses 48 million square kilometers of marine ecosystems across 132 nations (Froehlich et al., 2019; FAO, 2022). However, it is noteworthy that only 61 countries are currently engaged in seaweed cultivation, either commercially or at the experimental pilot stage (Froehlich et al., 2019; FAO, 2022). The interest in macroalgae production is correlated with their significant nutrient contents, including high carbohydrate levels (up to 60%), moderate to high protein levels (10–47%), and relatively low lipid content (1–3%), alongside varying mineral and ash content (7–38%) (Dominguez and Loret, 2019; Pappou et al., 2022). They are excellent sources of different vitamins (Ortiz et al., 2006; Besada et al., 2009; Hagan and Anyangwe, 2023), while also providing essential minerals like Ca, P, Na, K, and >54 trace elements (Dhargalkar and Pereira, 2005; Moreda-Piñeiro et al., 2012). Hence, the cultivation of seaweed under sustainable frameworks exhibits potential as a prospective approach to address the escalating need for worldwide food security and its multifaceted application across diverse sectors (Robin et al., 2018; Dominguez and Loret, 2019).

Ulva is a prevalent genus of green seaweed and is notable for its widespread distribution in brackish water and along shorelines in shallow waters (Morais et al., 2020). These organisms typically manifest in various forms, including slender threads, irregular sheets, or branching fronds, often with relatively large forms (Abudabos et al., 2013; Ghaderiadekani, 2019). *Ulva* species are known for their abundance of rare cell-wall polysaccharides, valuable sources of dietary fibers (Ale et al., 2011; Corino et al., 2019), enriched with iron, unsaturated fatty acids, and essential amino acids (Li et al., 2018; Dominguez and Loret, 2019). Among the *Ulva* species, sea lettuce (*Ulva lactuca*) can be found in a wide range of climates, from tropical to arctic, mostly in the upper to mid-intertidal zones (Abudabos et al., 2013; Carl et al., 2014). *U. lactuca* presents promising prospects for cultivation due to its global distribution (Kirkendale et al., 2013), high growth rates (Bruhn et al., 2011), and its capacity to adapt to various environmental conditions (Luo and Liu, 2011; Carl et al., 2014). Moreover, the demand for its polysaccharides in the pharmaceutical and food industries is rising due to their antiviral, antitumor, anticoagulant, anti-lipidic, hepatoprotective, immune-stimulating, antidepressant, and anxiolytic properties (Dominguez and Loret, 2019; Sari-Chmayssem et al., 2019; Morais et al., 2020).

At present, *U. lactuca* has either been collected from natural populations of shallow coastal areas (Cecchi et al., 1996; Bruhn et al., 2011) or grown in land-based systems, either alone or as part of integrated multitrophic aquaculture systems (Ryther et al., 1984; Neori et al., 1991; Msuya and Neori, 2008; Robertson-Andersson et al., 2008). *Ulva* species have been cultivated in several regions of the world in pilot commercial systems due to their potential utility as a food source, animal feed,

biofuel, and for bioremediation purposes (Gao, 2016). The initial experiments in commercial *Ulva* cultivation took place in America, where *Ulva* was utilized as biomass for the conversion of biomethane (Fannin, 1983; Gao, 2016). Research in Denmark focused on outdoor tank cultivation to explore the potential of *Ulva* in biomass production for bioenergy (Bruhn et al., 2011). *Ulva* collected from the integrated system was also utilized as animal fodder in South Africa (Robertson-Andersson et al., 2008; Bolton et al., 2009). Furthermore, *Ulva* has been deliberately grown as a biofilter in integrated multi-trophic systems in Israel (Neori et al., 2000) and Saudi Arabia (Al-Hafedh et al., 2015). Land-based *Ulva* cultivation has recently been carried out in Australia to study its potential in nutraceuticals and cosmeceuticals (Mata et al., 2016). Notwithstanding its considerable potential, sea lettuce cultivation encounters a number of obstacles, such as difficulty in maintaining a vegetative state, fluctuating growth rates, vulnerability to environmental stressors, and the requirement for optimized cultivation strategies (Ale et al., 2011; Bruhn et al., 2011; Pérez-Mayorga et al., 2011; Castelar et al., 2014).

Cultivation methods play a critical role in determining the productivity and nutritional quality of sea lettuce biomass. The floating cultivation method involves suspending sea lettuce in the water column using buoyant structures in the sub-tidal zones (Banik et al., 2023), while off-bottom cultivation entails securing seaweed to horizontal lines or nets positioned above the seafloor in the intertidal zones (Mohiuddin et al., 2023). Each cultivation method is influenced by factors such as water motion, nutrient availability, and light penetration, which in turn affect the growth and biochemical composition of sea lettuce (Messyasz and Rybak, 2011; Jin et al., 2018; Banik et al., 2023). However, the influence of different cultivation strategies on the production performances and biochemical composition of *U. lactuca* has not been intensively documented.

Variations in the productivity and biochemical composition of *Ulva* species have been found depending on site-specific ecological conditions like water quality, temperature, light exposure, and nutrient levels in water (Dawczynski et al., 2007; Manivannan et al., 2008; Messyasz and Rybak, 2011; Mišurcová, 2011; Jin et al., 2018; Morais et al., 2020). Seasonality has also been reported to have an impact on the growth conditions and biochemical contents of seaweed (Lamare and Wing, 2001; Bruhn et al., 2011). The winter season offers optimal environmental conditions for seaweed cultivation, leading to substantial development (Banik et al., 2023); conversely, the monsoon season results in low seaweed production (Ali et al., 2017; Banik et al., 2023). The environmental conditions of the cultivation site are crucial for obtaining optimal production, as they control the availability of nutrients and the rate of photosynthesis in seaweed (Hurd, 2000; Radiarta et al., 2014; Mohiuddin et al., 2023). The growth responses of algal flora in aquatic ecosystems are significantly influenced by the physio-chemical properties of water (Abdulwahab and Rabee, 2015; Sharma et al., 2016). For most seaweeds, the optimal temperature range falls between 22 and 30 °C (Budidaya, 2005). Typically, the growth rate of *Ulva* increases with rising temperatures but can decline when temperatures reach damaging levels (Ansari et al., 2011). The salinity range that is most favourable for the highest level of seaweed development is 24–35 ppt (WWF, 2014). According to the report by WWF (2014), the ideal pH range for seaweed growth is between 6 and 9, with a preference for 6.5 to 8.5. Transparency, influenced by turbidity, controls the amount of sunlight penetrating the water, making it a significant factor impacting seaweed growth as these plants rely on solar energy for photosynthesis (WWF, 2014). Besides these parameters, *Ulva* growth is also influenced by different nutrient concentrations, such as PO₄-P, NO₃-N, NO₂-N and NH₃-N (Oliveira et al., 2000; Masyahoro and Mappiratu, 2010). When the circumstances are favourable, the opportunistic genus *Ulva* can sustain rapid growth (Littler, 1980; Xia et al., 2004). As a result, it consists of highly productive species that frequently contribute a significant portion of the overall primary production in estuaries (Sfriso et al., 1992; Taylor et al., 2001; Xia et al., 2004). Although several

studies have been conducted to assess the biochemical properties of *U. lactuca* (Yaich et al., 2011; Nunes et al., 2017; Rasyid, 2017; Nissen et al., 2024), the influence of site-specific ecological factors on the production performances and biochemical composition of sea lettuce has not been documented comprehensively.

Bangladesh, situated in the northeastern Bay of Bengal, boasts a 710-km coastline and an extensive maritime area of about 119,000 km², with naturally growing seaweeds available between October and April along the southern coast (Islam and Aziz, 1987; Islam, 1998; Islam and Shamsuddoha, 2018). Out of the 335 species of seaweeds, 32 species are found in large numbers in marine settings (Billah et al., 2018; Siddiqui et al., 2019; Abdullah et al., 2020; Hossain et al., 2021), while species like *Gracilaria*, *Hypnea*, and *Ulva* exhibit adaptability to brackish water environments such as the Bakhkhali River estuary, Moheshkhali, and Kutubdia channels (Baweja et al., 2016; Sarker et al., 2021; Chowdhury et al., 2022). Seaweed aquaculture was initially established in Bangladesh in 2005 (Zafar, 2004, 2005, 2007) and further expanded in the following years (Hoq et al., 2016; Islam et al., 2017; Banik et al., 2023; Mohiuddin et al., 2023). Seaweed cultivation in Bangladesh mainly involves long-line, net, raft/hanging methods in the intertidal, lower intertidal, and subtidal zones (Hossain et al., 2020; Banik et al., 2023; Tasnim et al., 2024). Multiple studies have examined the growth performance of several seaweed species in different cultivation systems (Buschmann et al., 2008; Mantri et al., 2009; Sarkar et al., 2019; Banik et al., 2023; Mohiuddin et al., 2023). Despite the known benefits of *U. lactuca*, significant gaps remain in understanding how different cultivation systems affect its growth and biochemical contents (Nunes

et al., 2017; Jansen et al., 2022; Zollmann et al., 2023), particularly in the context of varying environmental conditions (Messyasz and Rybak, 2011; Jin et al., 2018). In this study, our objective is to evaluate the production performance of *U. lactuca* across three distinct cultivation systems, aiming to identify the most productive systems along with optimal cultivation locations, with a specific focus on the influence of site-specific ecological factors and seasonality. By evaluating the biochemical compositions of *U. lactuca*, we also aim to identify nutritional disparities and food safety aspects across different cultivation systems to improve seaweed mariculture sustainability.

2. Materials and methods

2.1. Study area

The *U. lactuca* experiment was conducted from October 2022 to June 2023 at three distinct coastal sites on the southeastern coast of Cox's Bazar, Bangladesh (Fig. 1). The study area encompassed Khurushkul (21° 30' 40.13" N, 91° 59' 53.3" E), Nuniarchora (21° 28' 26.6" N, 91° 57' 56.4" E), and Sonapara (21° 29' 53.54" N, 92° 04' 98.64" E). These locations were chosen due to their favourable water quality parameters (Zafar, 2005; Tasnim et al., 2024) and the presence of sand-flat intertidal zones, offering suitability for employing both semi-floating (off-bottom systems) and floating (floating long-line) cultivation systems (Banik et al., 2023; Mohiuddin et al., 2023).



Fig. 1. The locations of different cultivation systems of sea lettuce (*Ulva lactuca*) on the southeast coast of the Bay of Bengal, Bangladesh. The study was carried out in the Khurushkul, Nuniarchora, and Sonapara areas of Cox's Bazar district, Bangladesh. Depending on the site's suitability for different cultivation systems, 5 floating long-line cultivation (FLL) plots were established in the Khurushkul area; 5 floating long-line (FLL), 5 off-bottom long line (OBLL), and 5 off-bottom net (OBN) plots were established in the Nuniarchora area; and 5 off-bottom long line (OBLL) and 5 off-bottom net (OBN) plots were established in the Sonapara area. The OBLL and OBN cultivation plots were installed in the intertidal zone, and raft-based FLL cultivation plots were installed in the subtidal zone.

2.2. Construction and installation of seaweed culture systems

This study involved establishing three distinct cultivation methods: off-bottom net (OBN), off-bottom long-line (OBLL), and floating long-line (FLL) systems. The OBLL and OBN setups were placed on the sandy bottom within the intertidal zone, while the FLL system was positioned in the sub-tidal region with an average water depth of 3 m (Fig. 1). The off-bottom systems were partially submerged during high tide and rested on the exposed sand flat during low tide. Conversely, the FLL cultivation system stayed submerged in water throughout both high and low tides. The study involved setting up a total of 30 seaweed cultivation plots: 10 OFLL, 10 OBN, and 10 FLL systems. Depending on the site suitability for different cultivation systems, 15 plots (5 FLL, 5 OBLL, and 5 OBN) were established in Nuniachora, 10 plots (5 FLL, 5 OBLL) were established in Sonapara, and 5 plots (FLL) were established in Khurushkul of the Cox's Bazar district, Bangladesh (Fig. 1). OBLL plots were established by placing six bamboo poles parallel to the sandy mud bottom, with an additional set of five poles placed 10 m away, resulting in a 30 m² (10 m × 3 m) area for each plot (Fig. 2 A-C). These

poles were interconnected using polypropylene rope, with two plastic floats attached to each rope to keep them afloat during high tide. For the OBN plots (Fig. 2 D–F), a nylon rope-made net (5 m × 3 m) with a 15 cm mesh size was secured to bamboo poles on all sides, equipped with eight floats in the middle parts of the net to maintain buoyancy. The FLL plots (Fig. 2 G–I) were created by firmly securing ten recycled plastic drums underneath a raft built from two 10-m horizontal bamboo poles and five vertical 3-m bamboo poles, resulting in a 30 m² (10 m × 3 m) area. Six parallel long-line ropes with attached floats were installed on each raft, and the structure was anchored securely using two iron bar stones (Fig. 2 G–I).

2.3. Seed collection and inoculation in the culture systems

The initial *U. lactuca* seaweed specimens were collected from the Bangladesh Fisheries Research Institute. Before stocking, these specimens were washed with clean seawater to remove any debris, associated algae, or possible epiphytes. Fragments of the thalli were then utilized as planting stocks in tanks. Aeration was provided via a perforated pipe



Fig. 2. Design and photographic images of different cultivation systems and inoculation techniques for sea lettuce (*Ulva lactuca*). (A–C) Off-bottom long-line (OBLL) cultivation system with an area of 30 m² (10 m × 3 m) for a plot; (D–F) Off-bottom net (OBN) cultivation system with an area of 15 m² (5 m × 3 m) for a plot; (G–I) Floating long-line (FLL) cultivation system with an area of 30 m² (10 m × 3 m) for a plot; (J) Young and healthy growing thallus of *U. lactuca* as seed; (K–L), Inoculation techniques by placing young and healthy thallus of *U. lactuca* into holes created by untwisting and pulling the rope by the participatory farmers; and (M), An inoculated long line rope ready to use in FLL or OBLL cultivation systems.

positioned along the bottom of each tank, facilitating the suspension of *Ulva* thalli. Following one month of stocking, healthy, young, growing thalli were carefully selected as inoculum (Fig. 2J). For the OBLL and FLL systems, vegetative fragments of thalli were introduced as seeds by placing them into holes created by untwisting and pulling the rope (Fig. 2 K-M). Each 10 m long-line rope required around 300 g of seed for inoculation, with around 50 thalli fragments weighing around 6 g each. Thus, about 1.8 kg of seed was used for each OBLL and FLL cultivation plot (comprising 6 long-line ropes) (Figs. 2–3). In the OBN system, whole thalli fragments (approximately 5–6 g each) were placed in each knot joint, and a net plot required approximately 1.8 kg (300–360 knots per plot) of seed (Fig. 3).

2.4. Water quality parameters

During the peak high tide at each experimental site, physical and chemical variables were observed and recorded monthly for every cultivation plot. The physical parameters measured were surface water temperature, pH, dissolved oxygen (DO), salinity, and turbidity. The chemical parameters measured were phosphate-phosphorus ($\text{PO}_4\text{-P}$), nitrate-nitrogen ($\text{NO}_3\text{-N}$), nitrite-nitrogen ($\text{NO}_2\text{-N}$), and ammonia-nitrogen ($\text{NH}_3\text{-N}$). The temperature was measured with a Celsius thermometer, pH was measured using a digital pH meter (EcoSence pH10A by YSI), DO was monitored using a digital DO meter (Lutron PDO-519), salinity was determined using a Bellingham and Stanley *E-Line* Refractometer (Xylem), and turbidity was measured using a digital turbidity meter (Turb430 IR by WTW). The TSS of water samples was quantified using the APHA (2005), which involved filtering seawater with a



Fig. 3. The photographic views of the growing phase of sea lettuce (*Ulva lactuca*) cultivated in the three different systems with participatory farming approaches involving coastal communities along the southeast coast of the Bay of Bengal, Bangladesh. (A-D) Off-bottom long-line (OBLL) cultivation plots of *Ulva lactuca*; (E-F) Off-bottom net (OBN) cultivation plots of *Ulva lactuca*; (G-I) Floating long-line (FLL) cultivation plots of *Ulva lactuca*.

vacuum pump and Whatman filter paper.

2.5. Harvesting and growth performance estimations

Partial harvesting of cultured *U. lactuca* was done after it reached a length of 20–30 cm (Fig. 3). This process was conducted every 15 days during peak seasons (December–March) and every 30 days during off-peak seasons (October–November and March–April) (Padhi et al., 2011; Yang et al., 2006). Partial harvesting involved using a knife to cut the seaweed, leaving about 4–6 cm from the base to facilitate regrowth. The harvested *U. lactuca* from each plot was weighed using a digital scale (WPCS-DS758X, Walton, Bangladesh) and then sun-dried to determine its dry weight (Fig. 4).

The growth rate (%/day) and monthly biomass yield ($\text{kg}/\text{m}^2/\text{month}$) for each cultivation plot were calculated after each monthly harvest using specific formulas.

$$\text{GrossweightbiomassGWBkg}/\text{m}^2/\text{month}) = W_t/A$$

(Doty, 1986).

Where,

W_t = Gross weight at the end of the month (t),

A = Area of a culture plot (m^2).

$$\text{NetWeightBiomassNWBkg}/\text{m}^2/\text{month}) = (W_t - W_i)/A$$

(Doty, 1986).

Where,

W_0 = Initial seed weight at the beginning,

W_t = Weight at the end of the month (t),

A = Area of a culture plot (m^2).

$$\text{Daily Growth Rate (DGR; \%}/\text{day}) = \left[(W_t - W_0)^{1/t} \right] 100 / \text{day}$$

(Hung et al., 2009).

Where,

W_0 = Initial wet weight,

W_t = Final wet weight after day (t),

t = Duration of culture in days.

$$\text{Specific Growth Rate (SGR; \%}/\text{day}) = [(\ln W_t - \ln W_0)/t] 100 / \text{day}$$



Fig. 4. Photographic views of the harvesting, drying, and packaging of sea lettuce (*Ulva lactuca*) by the participating farmers. (A–B), Participated youth farmers showing the harvested *U. lactuca*; (C) Harvested and washed *U. lactuca*; (D–F), Sun drying of washed *Ulva lactuca*; (G–J) Packaging of dried *U. lactuca* for selling in the local market.

(Bhatia et al., 2015).

Where,

W_0 = Initial wet weight,

W_t = Final wet weight after day (t),

t = Duration of culture in days.

2.6. Sampling and sample preparation for biochemical analysis

During the peak growing month (February), approximately 1 kg of *U. lactuca* was manually harvested from each cultivation plot using knives from the Nuniarchora site, as this site was suitable for all three cultivation systems. The collected seaweed underwent thorough washing with seawater, repeated 2–3 times, to remove biofouling organisms and other impurities. In the laboratory, the samples were rinsed with clean water and then left to air-dry for one day. Following air drying, a part of the samples was freeze-dried for fatty acid profiling, whereas the remaining air-dried samples were further dried at 60 °C using a hot air oven until they reached a constant weight for proximate and mineral and heavy metal analysis. Subsequently, a blender machine (Kiam, BL-100, 750 watts, Bangladesh) was used to turn the dried samples into fine powder form. The powdered seaweed from 1 to 2 replicated plots was combined to create a representative replicated sample. Ultimately, three replicated samples were generated for each cultivation system from five plots.

2.7. Proximate composition analysis

The protein, lipid, fiber, and ash content were assessed in the samples collected from each culture system in Nuniarchora in February. Crude protein content was determined by the Kjeldahl method (Guebel et al., 1991) and calculated from the nitrogen content using a conversion ratio of 5.85 (AOAC, 2000). Total lipid content was analyzed by Soxhlet methods (AOAC, 2005). Fiber content was determined using Raw Fiber Extractors (Model: FIWE3, VELP Scientifica) and a muffle furnace that maintained samples at a temperature of 550 °C for 3 h. The ash content of *U. lactuca* was determined according to the incineration method (AOAC, 2000), using a muffle furnace at 550 °C for 5 h for the ash determination of the samples.

2.8. Fatty acids analysis

The analysis of fatty acids was performed following the approach described by Prato et al. (2017). The Soxhlet apparatus was initially employed to extract lipids from 2 g of dried seaweed samples. The lipid extraction method was conducted using diethyl ether as the solvent. The temperature was sustained at 60 °C during the final stage of lipid extraction. The fatty acid methyl esters (FAMES) were analyzed using the extracted lipid samples. The analysis was performed using gas chromatography-mass spectrophotometry (GC-MS) with a GCMS-QP2020 instrument (Shimadzu, Japan) that has a flame ionization detector. The FAMES were separated using a capillary column that was 30 m long, with an internal diameter of 0.25 mm, a film thickness of 0.15 μm, and a phase ratio of 250. Helium was used as the carrier gas, flowing at a rate of 1.42 ml per minute. The column temperature increased from 180 to 280 °C at a rate of 5 °C per minute and then remained constant at 280 °C. Fatty acid methyl esters (FAMES) were identified by comparing their retention times with a reference standard (FAME mix C8-C24; Sigma-Aldrich, Germany). The measurements were initially conducted in parts per million (ppm) and then translated into percentages (%) of all detected fatty acids.

2.9. Minerals and heavy metal analysis

The concentrations of nine minerals (calcium, sodium, zinc, potassium, iron, manganese, nickel, cobalt, and copper) and three hazardous heavy metals (chromium, lead, and cadmium) were measured using a

flame atomic absorption spectrophotometer (AAS). At first, powdered seaweed samples, weighing 0.50 g, were placed in a microwave-safe Teflon container that was both clean and dry to facilitate microwave digestion. Subsequently, 8 mm of nitric acid (HNO_3) was added to the samples, and the process of digestion was conducted at a temperature of 200 °C for a duration of 10 min. Following the filtration procedure, the digested samples were transferred into 50-ml volumetric flasks and subsequently diluted with distilled water. Then the transparent solutions were transferred into a falcon tube for examination using atomic absorption spectroscopy (AAS) (Model: AA-6300, Shimadzu, Kyoto, Japan). This study involved the use of single hollow cathode lamps for each element. A blank sample was processed similarly, without the inclusion of any seaweed sample. The mineral and heavy metal contents in seaweeds were measured using calibration curves that had coefficient values (r^2) > 0.998. The results were expressed in milligrams per kilogram (mg/kg) based on the weight of the sample after removing moisture. The assessment of the minerals and heavy metal concentrations in seaweeds was conducted utilizing quality assurance and quality control (QA/QC) techniques to ensure accuracy and precision.

2.10. Statistical analysis

The statistical analyses were conducted utilizing R, version 4.3.2 (R Development Core Team, 2023). The percentage data for statistical analysis were initially provided as percentages and subsequently transformed using the arcsine square root method. The Shapiro-Wilk test was used to assess the normal distribution assumptions of the datasets, while Levene's test was implemented to validate the homogeneity of variances using the 'onewaytests' package (Dag et al., 2018). Univariate analysis of variance (ANOVA) was performed using the 'car' package (Fox and Weisberg, 2019), followed by tukey multiple comparison tests utilizing the 'emmeans' package (Lenth Russell et al., 2022) to evaluate the comparative performances of different cultivation systems. The statistical significance of all effects was evaluated at a confidence level of 95%. The utilization of Performance Analytics packages (Peterson and Carl, 2020) was employed to evaluate and illustrate the associations between variables. Principal component analysis (PCA) was performed using the 'FactoMineR' package (Lé et al., 2008), focusing on the first and second principal components (PCs) as they accounted for the significant contribution to the overall variances in all scenarios. Graphs were generated using the 'ggplot2' software (Wickham, 2016), 'tidyverse' (Wickham et al., 2019), and 'ggside' (Landis, 2024) packages. The linear regression models were executed using the 'car' package (Fox and Weisberg, 2019).

3. Results

3.1. Variation of production performance of *U. lactuca*

The monthly variability of the GWB ($\text{kg/m}^2/\text{month}$), NWB ($\text{kg/m}^2/\text{month}$), and DGR (%/day) in three cultivation systems is presented in Fig. 5. The growth and production performance of *U. lactuca* fluctuated throughout the farming period. Combining the data from all three cultivation sites, the significantly highest ($p < 0.05$) growth and production performance parameters (GWB, NWB, and DGR) were observed in February, while the lowest was observed in June (Fig. 5). During the first harvesting in October, GWB, NWB, and DGR were comparatively low, reaching their highest value in February, and then showed a downward trend until reaching their valleys in June. For all the cultivation months, a one-way ANOVA test revealed that all the growth and production performance parameters were significantly higher in the FLL system than in the OBN and OBLL systems (Fig. 5). After June, *U. lactuca* did not survive anymore in any cultivation system at the cultivation sites.

The cultivation systems and site-specific variation in growth and production performance of *U. lactuca* are presented in Fig. 6. The GWB of

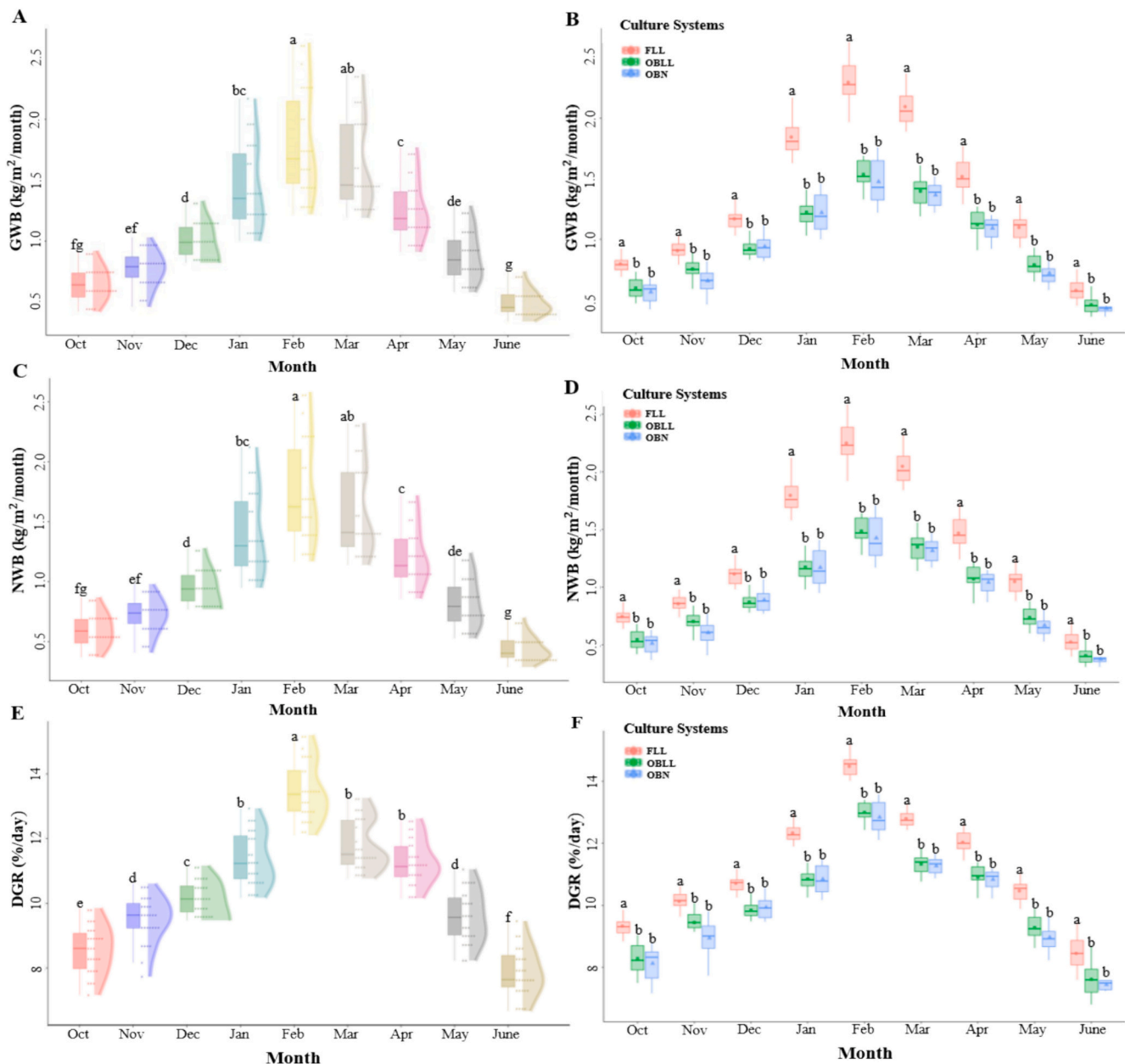


Fig. 5. Monthly growth and production performance of sea lettuce (*Ulva lactuca*) cultivated in the floating long-line (FLL), off-bottom long-line (OBLL), and off-bottom net (OBN) systems. (A) Monthly production of Gross Weight Biomass (GWB, kg/m²/month); (B) Monthly production of Gross Weight Biomass (GWB, kg/m²/month) in FLL, OBLL, and OBN cultivation systems; (C) Monthly production of Net Weight Biomass (NWB, kg/m²/month); (D) Monthly production of Net Weight Biomass (NWB, kg/m²/month) in FLL, OBLL, and OBN systems; (E) Daily growth rate (DGR, %/day) of *U. lactuca* on a monthly basis; (F) Daily growth rate (DGR, %/day) in FLL, OBLL, and OBN systems on a monthly basis. The raincloud plots and Box and Whiskers plots with the different superscript letters indicate significant differences ($p < 0.05$). Raincloud plots (A, C, and E) illustrate the variation in production across different months through a combination of density distribution representation and boxplots. The boxplots delineate median values (indicated by horizontal lines) and the upper and lower quartiles, with each dot symbolizing an individual sample. Error bars on these plots depict the maximum and minimum values. Conversely, boxplots (B, D, and F) elucidate the variation in production across different cultivation systems over various months to highlight mean values (marked by circles), medians (horizontal lines), and the upper and lower quartiles. Like the other plots, each dot represents a distinct sample, and error bars illustrate the range, from maximum to minimum. The analysis is based on the monthly production performance dataset of *U. lactuca* from 30 seaweed cultivation plots (10 OBLL, 10 OBN, and 10 FLL) for 9 months ($n = 270$).

U. lactuca in the FLL system was 45.38% and 39.85% higher ($p < 0.001$) than in the OBN and OBLL cultivation systems, respectively (Fig. 6A). Similarly, a 47.87% and 41.93% higher production ($p < 0.001$) of NWB was found in the FLL system compared to the OBN and OBLL cultivation systems, respectively (Fig. 6D). The DGR was found to be 12.92% and 11.22% higher ($p < 0.05$) in the FLL than in the OBN and the OBLL systems, respectively (Fig. 6G). The one-way ANOVA model also indicated significant variation for the GWB ($F = 20.8$, $p < 0.001$), NWB ($F = 20.795$, $p < 0.001$), and DGR ($F = 13.713$, $p < 0.001$) of *U. lactuca*

among three cultivation sites (Fig. 6). The GWB of *U. lactuca* in Khurushkul was 28.62% higher than the Nuniarchora and 59.76% higher than the Sonapara site (Fig. 6B). Similarly, NWB from Khurushkul was found to be 29.99% and 63.29% higher than the Nuniarchora and Sonapara sites, respectively (Fig. 6E). The average DGR in Khurushkul was 8.06% higher than the Nuniarchora site and 16.49% higher than the Sonapara site (Fig. 6H).

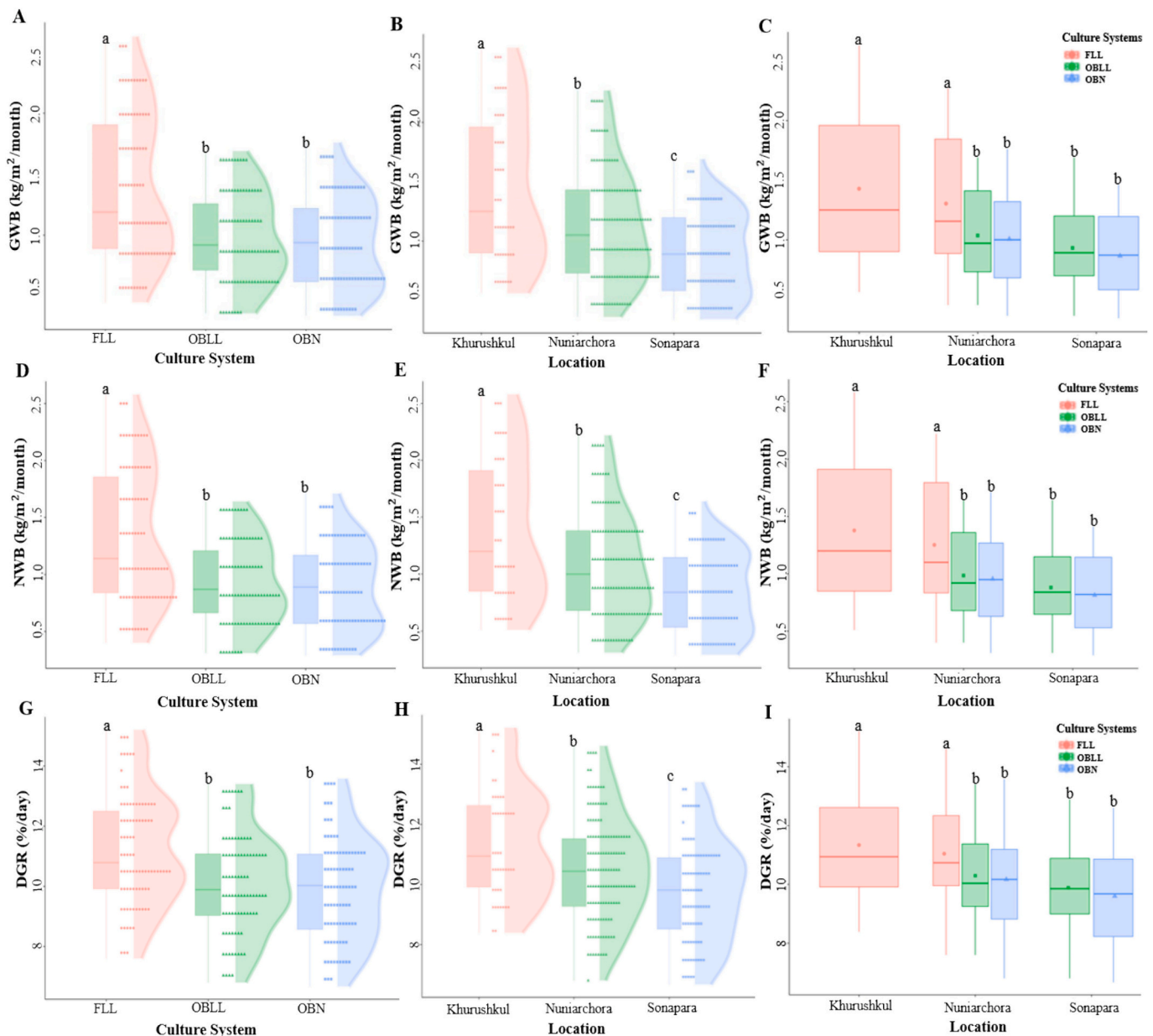


Fig. 6. Growth and production performances of sea lettuce (*Ulva lactuca*) in the floating long-line (FLL), off-bottom long-line (OBLL), and off-bottom net (OBN) cultivation systems in Khurushkul, Nuniarchora, and Sonapara locations. (A) Gross Weight Biomass (GWB, kg/m²/month) in different cultivation systems; (B) GWB (kg/m²/month) in different locations; (C) GWB (kg/m²/month) of *U. lactuca* in different cultivation systems and locations; (D) Net weight biomass (NWB, kg/m²/month) in different cultivation systems; (E) NWB (kg/m²/month) in different locations; (F) NWB (kg/m²/month) in different cultivation systems and locations; (G) Daily Growth Rate (DGR, %/day) in different cultivation systems; (H) DGR (%/day) in different locations; (I) DGR (%/day) in different cultivation systems and locations. The raincloud plots and Box and Whiskers plots with the different superscript letters indicate significant differences ($p < 0.05$). Raincloud plots (A, B, D, E, G, and H) illustrate the variation in production across different cultivation systems through a combination of density distribution representation and boxplots. The boxplots delineate median values (indicated by horizontal lines) and the upper and lower quartiles, with each dot symbolizing an individual sample. Error bars on these plots depict the maximum and minimum values of biomass. Conversely, boxplots (C, F, and I) elucidate the variation in production across different cultivation systems to highlight mean values (marked by circles), medians (horizontal lines), and the upper and lower quartiles. Like the other plots, each dot represents a distinct sample, and error bars illustrate the range of biomass, from maximum to minimum. The analysis is based on the monthly production performance dataset of *U. lactuca* from 30 seaweed cultivation plots (10 OBLL, 10 OBN, and 10 FLL) for 9 months ($n = 270$).

3.2. Ecological factor and growth performance of *U. lactuca*

Water quality parameters were estimated for three cultivation systems (Table 1). Among the parameters tested, significant variation was found in pH ($F = 4.806$, $p < 0.01$), DO ($F = 13.815$, $p < 0.001$), NO₃-N ($F = 8.731$, $p < 0.001$), and NH₃-N ($F = 20.237$, $p < 0.001$). It was found that DO and nutrient concentrations (NO₃-N and NH₃-N) were significantly higher in the FLL system compared to the OBN and OBLL systems.

Among the site-specific ecological parameters, pH ($F = 10.887$, $p < 0.001$), DO ($F = 78.401$, $p < 0.001$), NO₃-N ($F = 13.717$, $p < 0.001$), NH₃-N ($F = 1.794$, $p < 0.001$), and TSS ($F = 18.345$, $p < 0.001$) were found to vary among Khurushkul, Nuniarchora, and Sonapara cultivation sites (Table 2). Many of the water quality parameters were also found to vary significantly with seasonality and cultivation months (Supplementary Table 1).

The correlation analysis was also conducted between ecological

Table 1

Water quality parameters in different cultivation systems of sea lettuce (*Ulva lactuca*).

Water Quality Parameters	Cultivation Systems			F-Value	Sig. level p-value
	FLL	OBN	OBLL		
Temperature (°C)	27.70 ± 3.75	28.48 ± 3.57	28.40 ± 3.39	1.292	0.277 NS
Salinity (PPT)	29.87 ± 3.15	30.43 ± 3.36	30.15 ± 3.44	0.640	0.528 NS
pH	7.62 ± 0.37 ^b	7.78 ± 0.44 ^a	7.78 ± 0.42 ^a	4.806	0.009**
Turbidity (NTU)	61.57 ± 84.42	63.92 ± 89.35	64.56 ± 90.17	0.029	0.972 NS
DO (mg/l)	6.20 ± 0.96 ^a	5.67 ± 0.66 ^b	5.70 ± 0.62 ^b	13.815	0.000***
NO ₃ -N (mg/l)	0.97 ± 0.88 ^a	0.59 ± 0.72 ^b	0.55 ± 0.58 ^b	8.731	0.000***
NO ₂ -N (mg/l)	0.04 ± 0.03	0.03 ± 0.02	0.04 ± 0.07	0.557	0.574 NS
PO ₄ -P (mg/l)	0.38 ± 0.23	0.34 ± 0.15	0.34 ± 0.17	1.450	0.237 NS
NH ₃ -N (mg/l)	0.34 ± 0.15 ^a	0.26 ± 0.13 ^b	0.23 ± 0.06 ^b	20.237	0.000***
TSS (mg/l)	184.07 ± 160.27	186.05 ± 166.41	203.28 ± 183.31	0.347	0.707 NS

Here, FLL indicates floating long-line, OBN indicates off-bottom net, and OBLL indicates off-bottom long-line cultivation systems. All the values are represented as the mean ± SD of the 10 replicated cultivation units for each of the FLL, OBN, and OBLL systems. Different superscripts in the same row denote significant differences among different cultivation systems ($p < 0.05$).

Table 2

Water quality parameters in different cultivation sites of sea lettuce (*Ulva lactuca*).

Water Quality Parameters	Cultivation Locations			F-Value	Sig. level p-value
	Khurushkul	Nuniarchora	Sonapara		
Temp (°C)	27.62 ± 3.77	27.99 ± 3.85	28.79 ± 2.96	2.059	0.13 NS
Salinity (PPT)	29.39 ± 3.30	30.43 ± 2.94	30.12 ± 3.81	1.654	0.193 NS
pH	7.474 ± 0.33 ^b	7.79 ± 0.40 ^a	7.76 ± 0.43 ^a	10.887	0.000***
Turbidity (NTU)	57.86 ± 82.38	66.07 ± 89.85	62.02 ± 87.80	0.162	0.85 NS
DO (mg/l)	6.91 ± 0.44 ^a	5.55 ± 0.70 ^b	5.79 ± 0.61 ^b	78.401	0.000***
NO ₃ -N (mg/l)	0.99 ± 0.85 ^a	0.82 ± 0.88 ^a	0.39 ± 0.20 ^b	13.717	0.000***
NO ₂ -N (mg/l)	0.05 ± 0.04 ^a	0.04 ± 0.06 ^a	0.03 ± 0.02 ^a	3.201	0.042*
PO ₄ -P (mg/l)	0.36 ± 0.21	0.37 ± 0.20	0.32 ± 0.15	19.597	0.168 NS
NH ₃ -N (mg/l)	0.37 ± 0.17 ^a	0.27 ± 0.11 ^b	0.23 ± 0.10 ^b	1.794	0.000***
TSS (mg/l)	110.59 ± 49.59 ^b	248.96 ± 193.51 ^a	144.67 ± 138.13 ^b	18.345	0.000***

All the values are represented as the mean ± SD. Different superscripts in the same row denote significant differences among different cultivation sites ($p < 0.05$).

factors and the growth and production parameters of *U. lactuca* (Fig. 7). The correlation analysis revealed that salinity positively correlated with the biomass yield (GWB and NWB) ($r = 0.65$; $p < 0.001$) and growth rate (DGR and SGR) ($r = 0.71$; $p < 0.001$) of *U. lactuca*. The pH was also found to be positively correlated with DGR ($r = 0.40$; $p < 0.001$) and weight biomass ($r = 0.30$; $p < 0.001$) (Fig. 7). Similarly, different nutrients, including nitrate (NO₃-N), nitrite (NO₂-N), and phosphate (PO₄-P), were found to be positively correlated ($p < 0.001$) with the weight biomass and growth rate (Fig. 7). Temperature was found to be negatively correlated with GWB ($r = -0.59$; $p < 0.001$), NWB ($r = -0.59$; p

< 0.001), DGR ($r = -0.60$; $p < 0.001$), and SGR ($r = -0.60$; $p < 0.001$). Turbidity negatively influenced the growth rate ($r = -0.47$; $p < 0.001$) and biomass yield ($r = -0.43$; $p < 0.001$) of *U. lactuca* (Fig. 7). As expected, TSS had a negative correlation with the DGR ($r = -0.19$; $p < 0.01$), SGR ($r = -0.19$; $p < 0.01$), GWB ($r = -0.15$; $p < 0.05$), and NWB ($r = -0.15$; $p < 0.05$) of *Ulva lactuca* (Fig. 7). However, ammonia-nitrogen (NH₃-N) was not found to be significantly correlated ($p > 0.05$) with the growth rate and biomass production (Fig. 7).

A linear regression model was applied to explain the nature and magnitude of the relationships between water quality parameters and NWB of *U. lactuca* farmed in the FLL, OBLL, and OBN cultivation systems (Fig. 8). The linear regression lines for each cultivation system indicated the best-fit line that describes how NWB changes with water quality parameters within each cultivation system. The slope, intercept, regression coefficient (R^2), and significance levels (p) for all three cultivation plots are presented with equations (Fig. 8). For all three cultivation systems, the general linear regression model analysis showed a strong, significant positive relationship between NWB and salinity ($p < 0.001$; $R^2 = 0.51$ – 0.65) and NWB and NO₃-N ($p < 0.001$; $R^2 = 0.34$ – 0.39). The pH was found to have a significant positive relationship for the OBN ($p < 0.001$; $R^2 = 0.34$), OBLL ($p < 0.001$; $R^2 = 0.32$), and FLL ($p = 0.029$; $R^2 = 0.05$) cultivation systems (Fig. 8C). PO₄-P was also found to have a significant positive relationship with NWB for the OBLL ($R^2 = 0.10$, $p < 0.005$) and OBN ($R^2 = 0.045$, $p < 0.05$) systems but a non-significant relationship ($p > 0.05$) in the FLL system (Fig. 8F). For all three cultivation systems, a strong, significant negative relationship ($p < 0.001$) was observed between NWB and temperature (Fig. 8A) and NWB and turbidity (Fig. 8D). The other three water quality variables (DO, NO₂-N, and NH₃-N) did not show any significant relationship with the NWB of *U. lactuca* in any of the cultivation systems.

The multivariate PCA analysis was performed for a deep understanding of the interrelatedness among seasonality, ecological factors, and the production performance of *U. lactuca* seaweed (Fig. 9). The first two principal components (PC1 and PC2) cumulatively explained 63.1% of the variability (Table 3). PC1 (48.1%) was associated with the growth and production parameters (GWB, NWB, SGR, and DGR) of *U. lactuca* seaweed, which was found to be positively correlated with salinity, pH, PO₄-P, NO₃-N, NO₂-N, and NH₃-N, and negatively correlated with temperature, turbidity, and TSS (Table 3; Fig. 9). The PCA investigation further revealed that high concentrations of different nutrients (NO₃-N, NO₂-N, and PO₄-P) positively influenced the biomass yield and growth rate of *U. lactuca* seaweed from December to March (Fig. 9). However, higher temperatures, turbidity, and TSS lowered the growth rate and biomass yield of *U. lactuca* during the months from October to November and March to June (Fig. 9).

3.3. Proximate composition

The protein content of *U. lactuca* varied significantly ($F = 36.773$, $p < 0.001$) among the three cultivation systems (Table 4). The FLL and OBN systems had a significantly higher amount of protein than the OBLL system (Table 4). Similarly, lipid content also varied significantly ($F = 7.437$, $p < 0.05$) among the three cultivation systems. While *U. lactuca* from the FLL system had lower lipid content, a similar content was found between the OBN and OBLL systems (Table 4). No significant difference in crude fiber ($F = 2.122$, $p > 0.05$) and ash ($F = 1.587$, $p > 0.05$) content was observed among the three cultivation systems.

3.4. Fatty acids content

Depending on the cultivation systems, the saturated fatty acid (SAFA) content of *U. lactuca* ranged from 65% to 71.44% of total fatty acids (Table 5). The SAFA content was significantly higher ($F = 13.034$, $p < 0.01$) in the OBLL system. Among the SAFAs, tridecanoic acid ($F = 35.81$, $p < 0.001$), myristic acid ($F = 27.701$, $p < 0.001$), stearic acid ($F = 52.309$, $p < 0.001$), and arachidic acid ($F = 77.660$, $p < 0.001$) varied

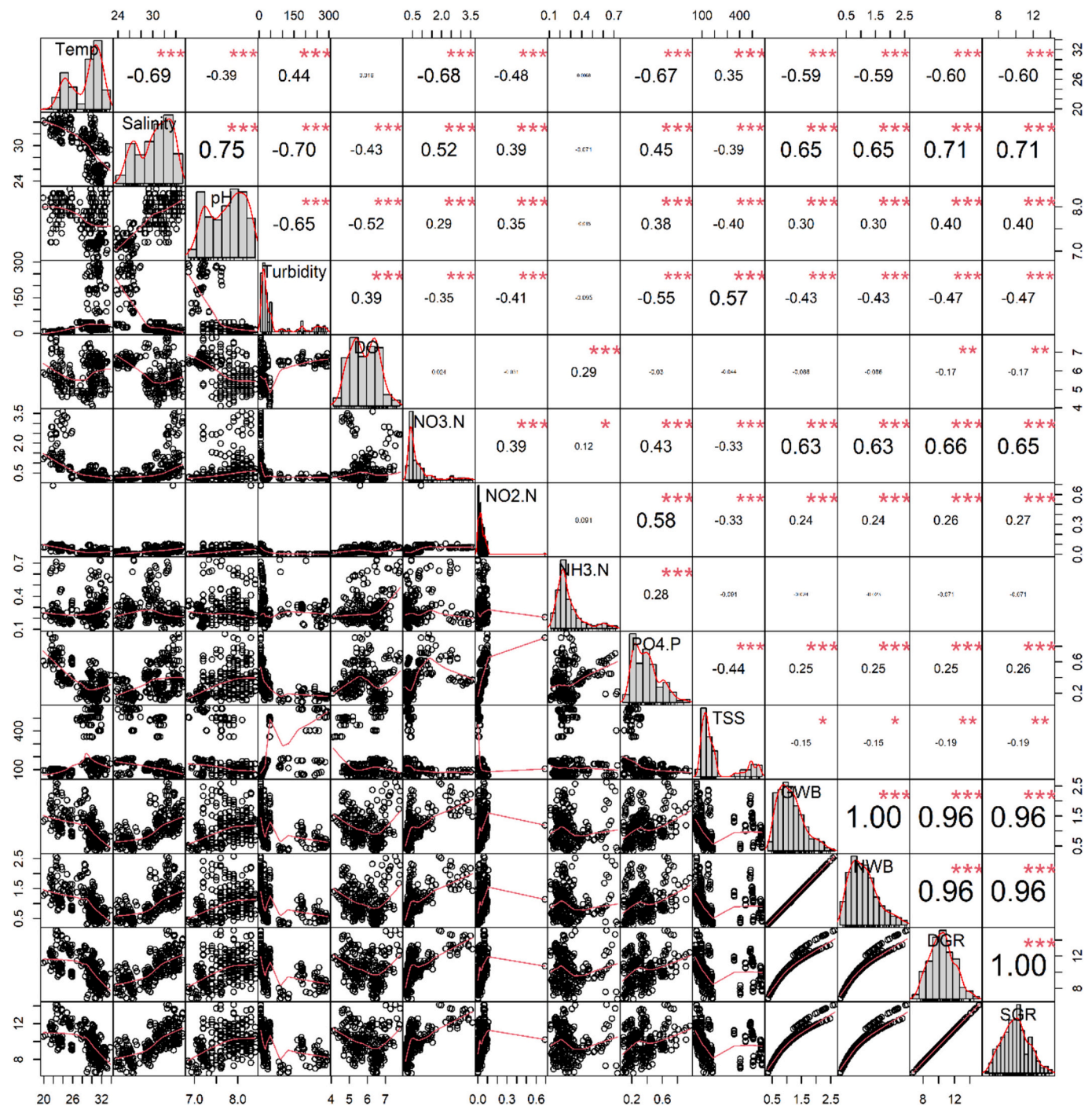


Fig. 7. Relationship among the water quality parameters and growth and production performance parameters of sea lettuce (*Ulva lactuca*) farmed in the floating long-line (FLL), off-bottom long-line (OBL), and off-bottom net (OBN) cultivation systems at the southeast coast of the Bay of Bengal, Bangladesh. Here, the variables' full forms: Temp- Water Temperature ($^{\circ}\text{C}$); Salinity (ppt); Turbidity (NTU); DO-Dissolved Oxygen (mg/l); $\text{NO}_3\text{-N}$ -Nitrate-nitrogen (mg/l); $\text{NO}_2\text{-N}$ -Nitrite-nitrogen (mg/l); $\text{PO}_4\text{-P}$ -Phosphate-phosphorus (mg/l); $\text{NH}_3\text{-N}$ -Ammonia-nitrogen (mg/l); TSS-Total Suspended Solid (mg/l); GWB-Gross Weight Biomass ($\text{kg}/\text{m}^2/\text{month}$); NWB-Net Weight Biomass ($\text{kg}/\text{m}^2/\text{month}$); DGR-Daily Growth Rate ($\%/ \text{day}$); SGR-Specific Growth Rate ($\%/ \text{day}$). The values given around all the axes are the range of each parameter's measured unit values. Correlation coefficients (r) are indicated with numeric values, where the font size of r gets bigger as the correlation gets stronger. The significance levels (p) are denoted by asterisks ($* < 0.05$, $** < 0.01$, $*** < 0.001$). The analysis is based on the monthly production performances of *U. lactuca* and the water quality dataset of 30 seaweed cultivation plots (10 OBL, 10 OBN, and 10 FLL) for 9 months ($n = 270$).

significantly (Table 5). The monounsaturated fatty acid (MUFA) contents of *U. lactuca* ranged from 6.2% to 9.46% of the total fatty acids (Table 5). The MUFA content was significantly higher ($F = 40.360$, $p < 0.001$) in the FLL and OBN systems compared to the OBL system. Among the MUFAs, palmitoleic acid ($F = 21.197$, $p < 0.01$), oleic acid ($F = 23.168$, $p < 0.01$), and erucic acid ($F = 533.265$, $p < 0.001$) varied

significantly among the three cultivation systems (Table 5). Polyunsaturated fatty acids (PUFA) ranged from 22.36% to 25.55% in *U. lactuca*, in which n3-PUFA accounted for 18.97% to 20.32% and n6-PUFA ranged from 2.5% to 6.5% (Table 5). Linolenic acid (C18:3n-3), eicosapentaenoic acid (C20:5n-3), and linoleic acid (C18:2n-6) were the major components of the PUFA contents in *U. lactuca* (Table 5). The n3-

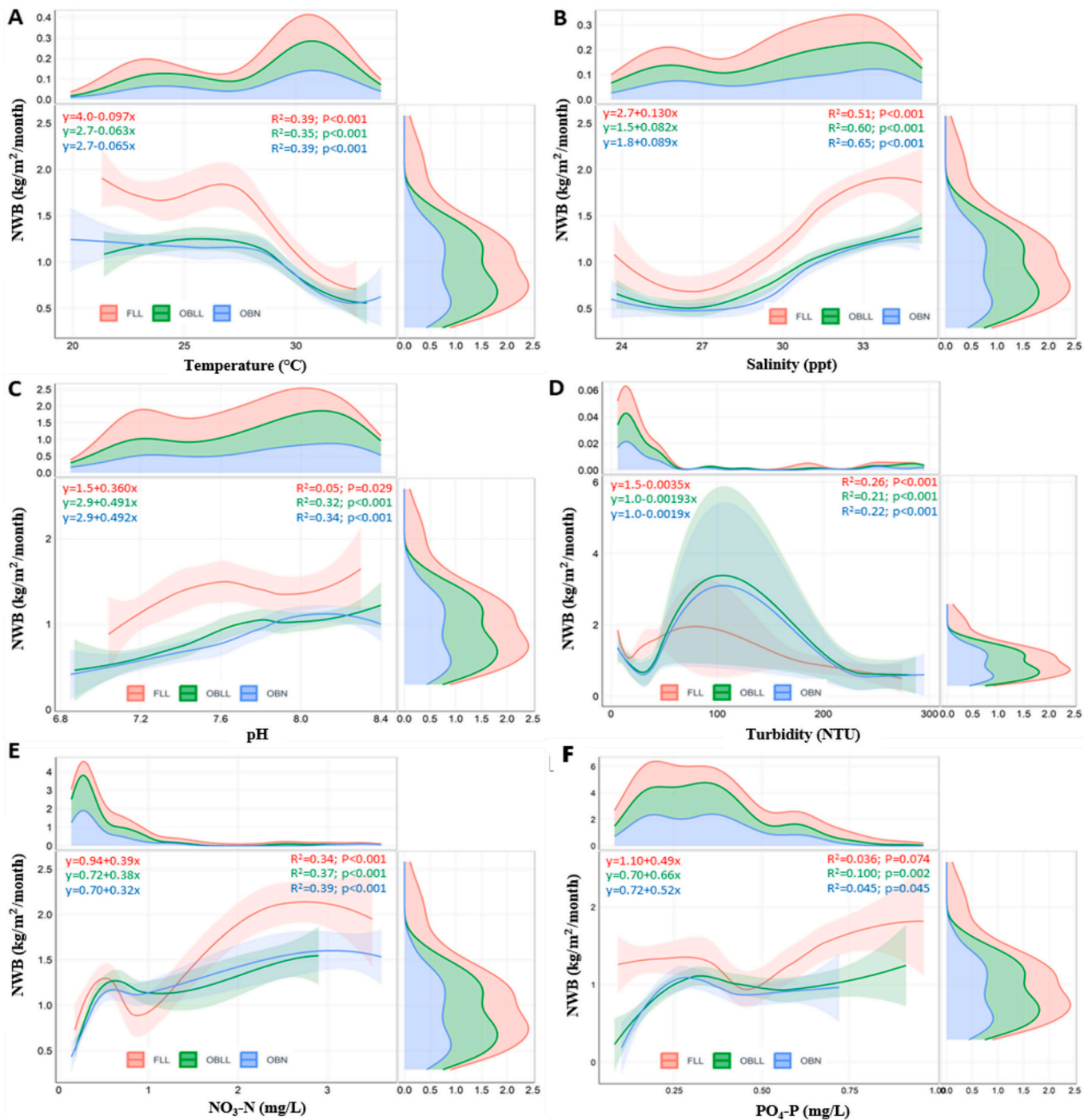


Fig. 8. The relationships between water quality parameters and net weight biomass (NWB) production of sea lettuce (*Ulva lactuca*) farmed in the floating long-line (FLL), off-bottom long-line (OBLL), and off-bottom net (OBN) cultivation systems at the southeast coast of the Bay of Bengal, Bangladesh. (A) Relationship between NWB (kg/m²/month) and temperature (°C); (B) Relationship between NWB (kg/m²/month) and salinity (ppt); (C) Relationship between NWB (kg/m²/month) and pH; (D) Relationship between NWB (kg/m²/month) and turbidity (NTU); (E) Relationship between NWB (kg/m²/month) and NO₃-N (mg/L); and (F) Relationship between NWB (kg/m²/month) and PO₄-P (mg/L). Each figure has two main components: a main plot (middle of the two marginal plots) and two marginal density plots (top-side and right-side of the main plot). In the main plot, biomass is put on the y-axis, and the water quality parameter is put on the x-axis. The plots are colored according to the cultivation systems: red for FLL, green for OBLL, and blue for OBN. The figure also shows linear regression lines for each cultivation system, which indicate the best-fit line that describes how biomass changes with water quality parameters within each cultivation system. The slope, intercept, regression coefficient (R^2), and significance levels (p) are presented with equations, which can be used to compare the biomass-water quality relationship across different cultivation systems. The marginal density plots show the distribution of biomass and water quality values across all observations. The top plot shows the density of measured water quality values on the x-axis, while the y-axis shows its frequency. The right plot shows the density of biomass values on the y-axis, while the x-axis shows its frequency. The density plots are useful for visualizing the shape, spread, and center of the data. The analysis is based on the monthly dataset of NWB and water quality parameters of 30 seaweed cultivation plots (10 OBLL, 10 OBN, and 10 FLL) for 9 months ($n = 270$). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

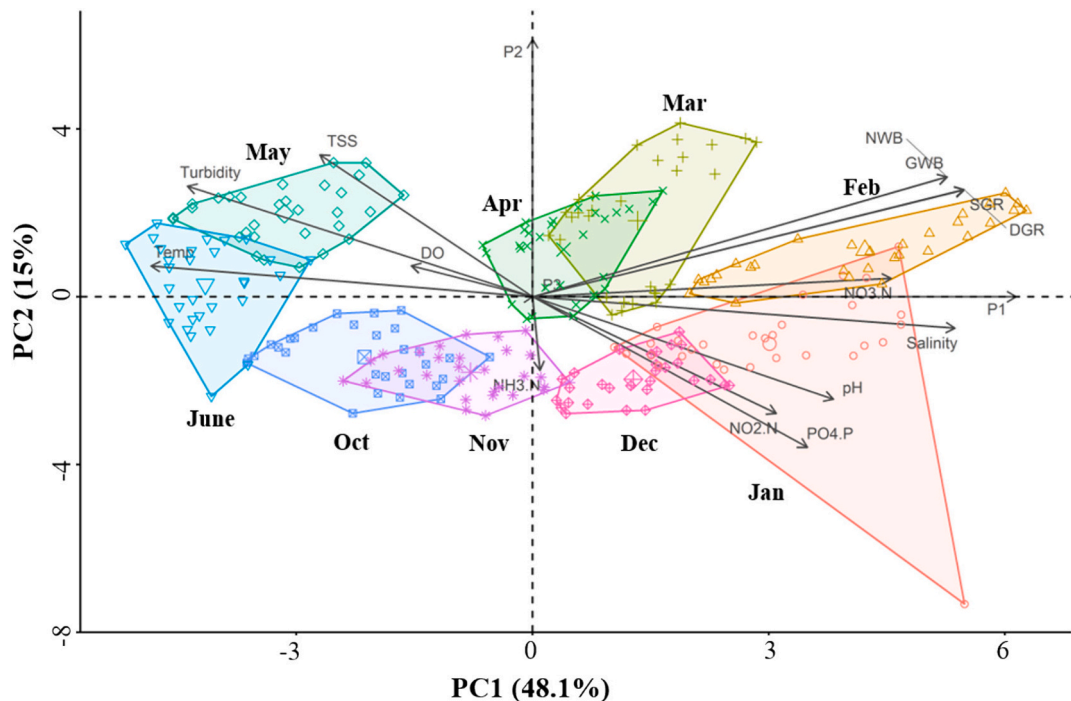


Fig. 9. Biplots of the principal component analysis (PCA) of the monthly variation of growth and production performance of sea lettuce (*Ulva lactuca*) with the water quality parameters of cultivation plots. Here, variables' full forms: Temp-Temperature (°C); Salinity (ppt); Turbidity (NTU); DO-Dissolved Oxygen (mg/l); NO₃-N-Nitrate-nitrogen (mg/l); NO₂-N-Nitrite-nitrogen (mg/l); PO₄-P-Phosphate-phosphorus (mg/l); TSS-Total Suspended Solids (mg/l); GWB-Gross Weight Biomass (kg/m²/month); NWB-Net Weight Biomass (kg/m²/month); DGR-Daily Growth Rate (%/day); SGR-Specific Growth Rate (%/day). The analysis is based on the monthly sampling dataset of production performances and water quality parameters of 30 seaweed cultivation plots (10 off-bottom long-line, 10 off-bottom net, and 10 floating long-line) for 9 months (n = 270).

Table 3
Principal Component Analysis (PCA) of the interrelation among the growth and production performance parameters of sea lettuce (*Ulva lactuca*) with the water quality parameters of the cultivation plots. Eigenvalues, explained, and cumulative loadings of the variables for the first five PCs.

Principal Components					
	PC1	PC2	PC3	PC4	PC5
Variance explained					
Eigenvalues	6.72849	2.10397	1.70997	0.87720	0.74854
% of variance	48.06064	15.02839	12.21408	6.26573	5.34677
Cumulative %	48.06064	63.08904	75.30312	81.56886	86.9156
Factor loadings					
Temperature	−0.78937	0.11971	−0.24392	0.32426	−0.09937
Salinity	0.87382	−0.12213	−0.28342	0.02202	0.01554
pH	0.62205	−0.39824	−0.47114	0.15498	0.03295
Turbidity	−0.71393	0.42924	0.23408	−0.22730	0.14465
DO	−0.25074	0.11991	0.84090	−0.07731	−0.27311
NO ₃ -N	0.74138	0.07131	0.33583	−0.09824	0.00317
NO ₂ -N	0.50390	−0.45316	0.21417	−0.34471	0.29911
NH ₃ -N	0.01514	−0.28605	0.55557	0.72542	0.24888
PO ₄ -P	0.56833	−0.58460	0.30072	−0.11554	0.22000
TSS	−0.43989	0.55159	−0.11447	−0.01171	0.66362
GWB	0.85732	0.46529	0.08755	0.08262	−0.01358
NWB	0.85704	0.46572	0.08755	0.08320	−0.01370
DGR	0.89223	0.41676	−0.00456	0.06299	−0.02554
SGR	0.89241	0.41489	−0.00722	0.06315	−0.02412

PUFA was not found to vary significantly ($F = 0.458$, $p > 0.05$) among the three cultivation systems. However, a significantly higher amount of n6-PUFA ($F = 20.86$, $p < 0.01$) was found in the FLL and OBN systems compared to the OBLL system.

Table 4
Variation of proximate composition (%) of sea lettuce (*Ulva lactuca*) under different cultivation systems.

Proximate composition	Cultivation systems			F-value	Sig. level p-value
	FLL	OBN	OBLL		
Protein (%)	19.251 ± 1.007 ^a	18.433 ± 0.440 ^a	16.841 ± 0.101 ^b	36.773	0.000***
Lipid (%)	0.130 ± 0.015 ^b	0.295 ± 0.056 ^a	0.298 ± 0.088 ^a	7.437	0.024*
Crude fiber (%)	13.090 ± 0.462	12.271 ± 1.047	16.806 ± 4.843	2.122	0.201NS
Ash (%)	10.995 ± 3.656	14.141 ± 2.372	14.943 ± 0.146	1.587	0.280NS

Here, FLL indicates floating long-line, OBN indicates off-bottom net, and OBLL indicates off-bottom long-line cultivation systems. All the values are represented as the mean ± SD of the five replicated cultivation units for each of the FLL, OBN, and OBLL systems. Different superscripts in the same row denote significant differences among different cultivation systems ($p < 0.05$).

3.5. Minerals and heavy metal content

The nine minerals (Ca, Na, K, Fe, Zn, Mn, Ni, Co, and Cu) and three heavy metals (Pb, Cd, and Cr) were measured for *U. lactuca* cultured in the three cultivation systems (Table 6). Ca was the most abundant mineral found in *U. lactuca*, followed by Na, K, and Fe (Table 6). Among the nine minerals analyzed, Na ($F = 7.397$, $p < 0.05$), Fe ($F = 7.188$, $p < 0.05$), and Co ($F = 6.097$, $p < 0.05$) varied significantly among the three cultivation systems (Table 6). Higher contents of Fe and Co ($p < 0.05$) were found in the OBLL system, whereas a higher content of Na was found in the FLL system (Table 6). Several trace minerals, such as Zn, Mn, Ni, and Cu, were also detected in an appreciated amount without any significant differences among the three cultivation systems. *U. lactuca* from the FLL system was found to have a comparatively lower

Table 5Variation of fatty acid contents (% of total fatty acid) of sea lettuce (*Ulva lactuca*) under different cultivation systems.

Carbon	Fatty acids	Cultivation systems			F-Value	Sig. level p-value
		FLL	OBN	OBLL		
C8:0	Octanoic acid	2.822 ± 0.227	3.667 ± 0.776	3.436 ± 0.459	1.985	0.218NS
C10:0	Decanoic acid	0.028 ± 0.004 ^b	0.042 ± 0.002 ^a	0.036 ± 0.004 ^{ab}	9.779	0.013*
C12:0	Lauric acid	10.345 ± 0.928	10.972 ± 0.758	10.905 ± 0.915	0.469	0.647NS
C13:0	Tridecanoic acid	1.455 ± 0.094 ^b	2.302 ± 0.088 ^a	1.735 ± 0.173 ^b	35.81	0.000***
C14:0	Myristic acid	0.148 ± 0.007 ^b	0.075 ± 0.017 ^c	0.231 ± 0.04 ^a	27.702	0.001**
C16:0	Palmitic acid	13.527 ± 0.991	12.567 ± 1.631	11.290 ± 0.814	2.631	0.151NS
C18:0	Stearic acid	26.006 ± 1.897 ^b	23.036 ± 2.626 ^b	37.836 ± 0.217 ^a	52.309	0.000***
C20:0	Arachidic acid	4.931 ± 0.278 ^a	4.74 ± 0.89 ^a	0.091 ± 0.021 ^b	77.66	0.000***
C17:0	Heptadecanoic acid	0.295 ± 0.139	0.433 ± 0.237	0.342 ± 0.037	0.581	0.588NS
C21:0	Heneicosanoic acid	5.246 ± 0.649 ^b	7.866 ± 0.380 ^a	5.414 ± 1.415 ^b	7.538	0.023*
C22:0	Behenic acid	0.177 ± 0.030	0.185 ± 0.016	0.113 ± 0.057	3.091	0.119NS
	ΣSAFA	64.988 ± 1.297 ^b	65.894 ± 2.333 ^b	71.437 ± 1.134 ^a	13.034	0.007**
C16:1	Palmitoleic acid	7.011 ± 0.523 ^b	8.208 ± 0.505 ^a	5.945 ± 0.122 ^b	21.197	0.002**
C18:1	Oleic acid	0.206 ± 0.023 ^b	0.592 ± 0.172 ^a	0.049 ± 0.006 ^b	23.168	0.002**
C20:1	Cis-11-Eicosenoic acid	0.12 ± 0.128	0.023 ± 0.006	0.022 ± 0.015	1.688	0.262NS
C22:1	Erucic acid	1.292 ± 0.083 ^a	0.012 ± 0.006 ^b	0.098 ± 0.039 ^b	533.265	0.000***
C24:1	Nervonic acid	0.832 ± 0.491 ^a	0.076 ± 0.074 ^b	0.083 ± 0.037 ^b	6.835	0.028*
	ΣMUFA	9.464 ± 0.4222 ^a	8.915 ± 0.701 ^a	6.200 ± 0.100 ^b	40.360	0.000***
C18:2n-6	Linoleic acid	4.800 ± 0.978 ^a	3.772 ± 0.820 ^a	1.005 ± 0.266 ^b	20.358	0.002**
C20:3n-6	Eicosatrienoic acid	1.218 ± 0.953	0.440 ± 0.048	0.781 ± 0.305	1.362	0.325NS
C20:4n-6	Arachidonic acid	0.550 ± 0.158	0.645 ± 0.345	0.769 ± 0.276	0.492	0.634NS
	Σn6-PUFA	6.569 ± 1.086 ^a	4.859 ± 0.438 ^a	2.556 ± 0.613 ^b	20.864	0.002**
C18:3n-3	Linolenic acid	7.079 ± 1.169 ^a	9.819 ± 1.639 ^a	9.742 ± 0.282 ^a	5.297	0.047*
C20:5n-3	Eicosapentanoic acid	11.201 ± 0.913	9.686 ± 1.422	9.647 ± 0.622	2.176	0.195NS
C22:6n-3	Docosahexaenoic acid	0.676 ± 0.118	0.735 ± 0.233	0.378 ± 0.017	4.796	0.057NS
C22:5n-3	Docosapentaenoic acid	0.018 ± 0.005	0.086 ± 0.087	0.033 ± 0.033	1.303	0.339NS
	Σn3-PUFA	18.976 ± 2.105	20.329 ± 1.959	19.803 ± 0.924	0.458	0.653NS
	ΣPUFA	25.547 ± 1.410	25.189 ± 2.178	22.360 ± 1.198	3.357	0.105NS
	Σn3/ Σn6	2.973 ± 0.785 ^b	4.196 ± 0.435 ^b	8.004 ± 1.679 ^a	17.078	0.003**
	ΣTUFA	35.011 ± 1.297 ^a	34.104 ± 2.333 ^a	28.561 ± 1.134 ^b	13.034	0.007**
	ΣSAFA/ ΣTUFA	1.858 ± 0.104 ^b	1.94 ± 0.202 ^b	2.504 ± 0.142 ^a	15.407	0.004**
	ΣSAFA/ ΣTFA	0.649 ± 0.013 ^b	0.658 ± 0.023 ^b	0.714 ± 0.011 ^a	12.632	0.007**
	ΣTUFA/ ΣTFA	0.349 ± 0.013 ^a	0.34 ± 0.023 ^a	0.285 ± 0.011 ^b	12.632	0.007**

Here, FLL indicates floating long-line, OBN indicates off-bottom net, and OBLL indicates off-bottom long-line cultivation systems. All the values are represented as the mean ± SD of the three replicated cultivation units for each of the FLL, OBN, and OBLL systems. Different superscripts in the same row denote significant differences among different cultivation systems ($p < 0.05$).

content of hazardous heavy metals than the off-bottom systems (Table 6). A significantly higher amount of Cr ($F = 11.555$, $p < 0.05$) and pb ($F = 5.752$, $p < 0.05$) were found in the OBLL system compared to the FLL system (Table 6).

4. Discussion

4.1. Production and growth performance of *U. lactuca* in different cultivation systems

The growth and production performance of *U. lactuca* were significantly higher in the FLL cultivation system compared to the OBN and OBLL systems (see Fig. 6). According to other research findings, the floating raft culture method in sub-tidal zones demonstrated superiority over all other currently available methods for cultivating seaweeds (Ganesan et al., 2011; Banik et al., 2023). Floating systems involve keeping seaweeds floating on the water's surface, typically established in subtidal areas, where a constant water flow removes mud and biota fouling from seaweeds, enabling the absorption of additional growth-stimulating nutrients (Putra and Aryawati, 2011; Banik et al., 2023). The floating technique enhances the gas exchange between seaweed and water by reducing sedimentation and self-shading while maximizing the seaweed's exposure to sunlight (Sobuj et al., 2023). Conversely, off-bottom culture systems in the intertidal zone undergo desiccation stress due to tidal changes, impacting nutrient absorption, photosynthesis rate (Radiarta et al., 2014), and seaweed growth (Hurd et al., 2014; Wang et al., 2016; Banik et al., 2023). During low tide, wind-induced evaporation of moisture exposes surface seaweeds to air (Ullah et al., 2023), potentially reducing the growth rate of certain

species under varying salinity conditions due to periodic dehydration (Li et al., 2021; Banik et al., 2023). Studies on *U. reticulata* cultivated in fishponds have shown biomass reduction due to desiccation (Msuya and Neori, 2002; Ullah et al., 2023). Off-bottom culture systems were positioned close to sediment and remained submerged, resulting in increased turbidity and reduced light penetration compared to the floating raft-based setup. The presence of suspended solid particles in water and the ability of light to penetrate the water have been found to be significant factors in restricting the growth of seaweeds (Ullah et al., 2023; Banik et al., 2023). Water transparency significantly affects light availability for photosynthesis (Kamlasi, 2008; Yuniarsih et al., 2015; Ullah et al., 2023). Light intensity does not influence photosynthetic performance but increases nitrate reductase activity, resulting in enhanced growth with less protein per gram of seaweed under higher light intensity (Yıldız and Tiryaki, 2017; Kraan, 2018). Decreased light intensity due to high turbidity in off-bottom systems directly correlates with reduced seaweed growth (Bunsom and Prathep, 2012; Ullah et al., 2023). The observed difference in production performance of *U. lactuca* might also be related with variation of different ecological factors. In our analysis, we evaluated the impact of ten water quality parameters on the net wet biomass (NWB) production of *U. lactuca* in three cultivation systems. Although six of these parameters demonstrated a statistically significant impact, the R^2 values of our models were relatively low, indicating that a large proportion of the variance in NWB production remains unexplained. This limitation highlights the complexity of *U. lactuca* growth dynamics and suggests the presence of other biotic and abiotic factors that were not accounted for in our study. Future studies should consider exploring other potential factors influencing *U. lactuca* growth, more precise measurements of light availability, and the

Table 6

Variation of heavy metals and minerals contents (mg/kg) of sea lettuce (*Ulva lactuca*) under different cultivation systems.

Minerals (mg/kg)	Cultivation systems			F- Value	Sig. level p-value
	FLL	OBN	OBLL		
Pb	2.313 ± 1.448 ^b	5.443 ± 1.307 ^b	10.717 ± 2.475 ^a	16.348	0.004**
Cd	0.910 ± 0.072	1.116 ± 0.125	1.180 ± 0.199	2.954	0.128NS
Cr	4.830 ± 1.575 ^c	9.490 ± 0.927 ^b	14.833 ± 1.901 ^a	32.292	0.001**
Co	1.073 ± 0.404 ^b	1.940 ± 0.877 ^{ab}	2.893 ± 0.539 ^a	6.097	0.036*
Ni	6.933 ± 0.380	9.253 ± 4.806	13.646 ± 3.786	2.784	0.14NS
Cu	15.966 ± 8.477	11.326 ± 1.618	21.100 ± 9.180	1.355	0.327NS
Zn	50.100 ± 14.6	58.016 ± 8.502	65.933 ± 2.713	1.926	0.226NS
Fe	360.700 ± 14.873 ^b	386.200 ± 14.900 ^{ab}	399.533 ± 6.653 ^a	7.188	0.026*
Mn	9.596 ± 0.876	10.508 ± 0.841	12.318 ± 2.923	1.725	0.256NS
Na	2603.789 ± 179.034 ^a	2292.500 ± 32.627 ^{ab}	2149.027 ± 180.678 ^b	7.397	0.024*
Ca	4910.600 ± 899.966	3769.700 ± 561.648	4792.467 ± 503.977	2.568	0.156NS
K	1389.367 ± 95.267 ^a	1070.224 ± 80.575 ^b	1118.100 ± 115.084 ^b	9.264	0.015*

Here, FLL indicates floating long-line, OBN indicates off-bottom net, and OBLL indicates off-bottom long-line cultivation systems. All the values are represented as the mean ± SD of the five replicated cultivation units for each of the FLL, OBN, and OBLL systems. Different superscripts in the same row denote significant differences among different cultivation systems ($p < 0.05$).

potential impacts of other macro and micronutrients.

Comparing the off-bottom culture systems, the production performance of *U. lactuca* was slightly higher in the OBLL system compared to the OBN cultivation system (see Fig. 6). The mesh-like dense structure in the OBN method might have constrained growth performance, likely due to the seaweed facing pressure as it competes for food, space, and nutrients (Wu et al., 2018; Ullah et al., 2023). Seaweed serves as the main food source for sessile, filter-feeding invertebrates such as acorn barnacles and mussels (Foster, 2018; Vinagre et al., 2020). Water close to sediment has an abundance of biofouling organisms and tiny particles that adhere to seaweeds and impede the absorption of nutrients and the process of photosynthesis (Ullah et al., 2023). The development of microorganisms on the surface of macroalgae reduces access to light and gases and alters interactions with other fouling epibionts, consumers, and pathogens (Goecke et al., 2010; Wahl et al., 2012; Lalegerie et al., 2020). Substrata-like net materials are more stable and facilitate rapid macrofouling settlement, which may impact the growth of *U. lactuca* in this study (Vinagre et al., 2020). Another reason for the comparatively lower production in the OBN system might be related to the initial inoculation density of *U. lactuca* seeds. The initial inoculation density was almost double in the OBN system (1.8 kg per 15 m²) than in the FLL and OBLL cultivation systems (1.8 kg per 30 m²). Previous studies indicated that seedling density and distance can significantly influence the development of seaweed (Sobuj et al., 2023; Ullah et al., 2023). Better water circulation in a less dense seedling inoculation system facilitates the dispersion of nutrients and promotes propagation, leading to enhanced growth rates of seaweed (Neish, 2008; Sobuj et al., 2023).

4.2. Production performance of *U. lactuca* in different locations

The growth responses of algal flora in aquatic ecosystems are influenced by the physio-chemical features (Abdulwahab and Rabee, 2015; Sharma et al., 2016) along with several biotic and abiotic elements in the aquatic environment (El-Shafai et al., 2007; Ansari and Khan, 2009; Lu

et al., 2010; Ansari and Ghanem, 2017). The observed difference of production performances in a particular area (see Fig. 6) is found to be influenced by the magnitude of environmental factors over time (Harley et al., 2012; Campbell et al., 2019; Mohiuddin et al., 2023; Tasnim et al., 2024). Differing production rates suggest that distinct *Ulva* spp. populations may adapt to specific local environmental conditions (Jansen et al., 2022). In cultivation experiments with sugar kelp *Saccharina latissima* at five different sites in Limfjorden, Denmark, biomass yield varied significantly, indicating a tenfold difference depending on the varying environmental conditions (Bruhn et al., 2016). Variations in physical and chemical parameters in the three respective locations were significant. Among the physical parameters, temperature, pH, turbidity, and total suspended solids were found lower in Khurushkul compared to the other two locations (see Table 2). Turbidity directly affects light penetration, thus limiting photosynthesis (Marinho-Soriano et al., 2009) and seaweed development (Ullah et al., 2023). Ideal temperature ranges (20 to 28 °C) promote nutrient absorption in seaweeds, resulting in faster development (Ullah et al., 2023), which is consistent with our result (see Table 2).

Alongside physical attributes, the presence of various nutrients also significantly influences the growth rate of seaweeds (Masyahoro and Mappiratu, 2010; Banik et al., 2023; Tasnim et al., 2024). *Ulva* species have a notable ability to assimilate nitrogen (NO₃-N, NO₂-N, and NH₃-N), particularly at high external concentrations, to fulfil their heightened nutrient demand and support rapid growth (Martínez et al., 2012; Roleda and Hurd, 2019). Higher nutrient concentrations observed in the Khurushkul area likely contributed to the accelerated growth rate of *U. lactuca* in that region (see Table 2). Rafts deployed in subtidal regions have better access to sunlight and higher water flow rates compared to intertidal regions. In Nuniachora and Sonapara, the lower flow rate in the off-bottom systems (Banik et al., 2023; Mohiuddin et al., 2023; Tasnim et al., 2024) might be the reason behind the comparatively lower production of *U. lactuca*. Some studies (Doty, 1971; Parker, 1981; Msuya and Neori, 2008; Ale et al., 2011) have provided evidence that the use of simulated current consistently improved the growth rates of *U. lactuca* in laboratory settings. Current velocities influence nutrient availability, with moderate flow rates potentially offering more nutrients for growth over time (Groenendijk et al., 2016). However, in our study, we did not assess the water flow rates in the respective culture locations; there are possibilities to further explore their impacts on *U. lactuca* growth.

4.3. Seasonality, environmental parameters, and production performance of *U. lactuca*

The current study revealed a distinct pattern in the seasonal variation of the biomass yield and growth of *U. lactuca*. The growth rates were found to be highest during the winter season (November–February), followed by a subsequent fall from March to June (see Fig. 5). The productivity of various *Ulva* species exhibited a pronounced seasonal fluctuation, as the growth of seaweed is influenced by external factors (Jansen et al., 2022). Previous studies have shown that the most favourable time for growing seaweed in the open coastal waters of Bangladesh is usually limited to a period of around five months, from November to March (Islam et al., 2019; Hossain et al., 2021; Mohiuddin et al., 2023). In the winter season (November–February), the low precipitation rate and the rapid evaporation of water greatly raise the concentration of different water nutrients in the Bay of Bengal (Asaduzzaman et al., 2019; Noor et al., 2021; Mohiuddin et al., 2023). Our study demonstrates that the presence of diverse nutrients is strongly and positively associated with the development rate and biomass of the *U. lactuca* seaweed (see Figs. 7–9). The presence of various nutrients such as nitrate (NO₃-N), phosphate (PO₄-P), and ammonia are crucial nutrients for seaweed growth as they enhance fertility and support primary productivity (Ovalle et al., 1999; Oliveira et al., 2000; Masyahoro and Mappiratu, 2010), which aligns with the findings of this study. Among the parameters, pH, dissolved oxygen (DO), nitrate-nitrogen

(NO₃-N), and ammonia-nitrogen (NH₃-N) significantly influenced the growth performance of *U. lactuca* (Table 1). During the pre-monsoon (March to June) and monsoon (July to October) seasons, the coastal regions along the Bay of Bengal receive substantial nutrient input, including nitrate, nitrite, phosphate, and ammonia (Banik et al., 2023). However, due to heavy rainfall and freshwater runoff, the overall concentration of these nutrients remains relatively low (Asaduzzaman et al., 2019; Noor et al., 2021). The increased turbidity and presence of suspended solids in the water resulting from freshwater runoff are likely to hinder seaweed growth, particularly during the months of October, March, and April (Banik et al., 2023).

In our study, temperature fluctuation between 22 and 32 °C exhibited a negative correlation with seaweed growth (see Fig. 7), hindering enzymatic processes (Piazzi et al., 2001) and reducing growth rates (Banik et al., 2023), notably impacting seaweed production in October–November and March–April (see Fig. 5). Similar temperature effects were noted in the growth of *Gracilaria* spp., leading to their inability to survive post-April when temperatures exceeded 30 °C (Banik et al., 2023). Previous research has indicated that seaweed development is enhanced in alkaline water with a pH range of 7–8.5 (Wisnu Ariyati et al., 2016; Banik et al., 2023). Seaweed growth was also found optimal with a pH range of 7.4–7.8. Higher turbidity and TSS levels during March and April, attributed to rainfall-induced soil erosion (Asaduzzaman et al., 2019, 2020; Noor et al., 2021), likely contributed to decreased seaweed growth and production during these periods (Banik et al., 2023). Salinity, ranging from 29 to 34 ppt during low rainfall and reduced surface runoff from December to February, potentially contributed to heightened seaweed growth rates during these months (Banik et al., 2023).

4.4. Influences of the culture systems on the biochemical composition of *U. lactuca*

The observed difference in different biochemical compositions of *U. lactuca* in different cultivation systems might be due to geographical differences and environmental conditions (Lourenço et al., 2002; Mišurcová, 2011). Seaweeds undergo constant changes during low tide, exposure to light, and air temperature, whereas submersion and various aquatic environment factors influence them during high tide (Cardozo et al., 2008; Guaratini et al., 2009; Ansari and Ghanem, 2017). As the FLL system undergoes minimum tidal impact and has maximum sunlight exposure, this may potentially boost photosynthesis rates, leading to higher protein and fatty acid content (Klyachko-Gurvich et al., 1999; Banik et al., 2023). Conversely, off-bottom systems, exposed to regular tidal fluctuations and higher air temperatures during low tide, experienced distinct conditions (Banik et al., 2023). Tidal fluctuations significantly impact light exposure and temperature, inducing oxidative stresses (Ansari and Ghanem, 2017) that impact the structural integrity of biomolecules, leading to protein oxidation, nucleic acid damage, and lipid peroxidation (Rezayian et al., 2019; Banik et al., 2023). To cope with thermal stress, seaweed changes protein synthesis patterns to maintain cellular homeostasis (Eggert, 2012). Sustainable seaweed cultivation as a dietary protein source can potentially lessen dependence on fishing by encouraging seaweed farming and utilizing culture techniques to boost the production of high-value seaweeds (Reddy et al., 2008; Brien et al., 2022).

Discrepancies in lipid content could be attributed to regional climate and geographical conditions influencing seaweed development (Ortiz et al., 2006). Seaweed adjusts to increased temperatures by reducing the concentration of its polyunsaturated fatty acids to maintain fluidity (Britton et al., 2021) in the cell membrane, perhaps leading to changes in its fatty acid compositions in off-bottom systems (Ortiz et al., 2006; Mišurcová, 2011; Britton et al., 2021; Banik et al., 2023). Specific fatty acids play a crucial role in the composition of dietary fats, significantly impacting human nutrition and health (Kumar et al., 2011). Reduced intake of saturated fatty acids and higher ratios of polyunsaturated fatty

acids to saturated fatty acids are strongly associated with decreased risks of coronary heart disease, hypertension, diabetes, and autoimmune diseases (Ajayan et al., 2021). Previous studies consistently demonstrate that the nutrient composition and metabolic activity of various seaweed species respond to ecological growth conditions (Benjama and Masniyom, 2012; Nazni and Deepa, 2015; Banik et al., 2023). Seaweeds alter their biochemical composition, including the quantities of sugars, amino acids, fatty acids, and minerals, in response to variations in ecological parameters (Stengel et al., 2011; Banik et al., 2023). However, the reasons for comparable protein and fatty acid content between the OBN and FLL systems remained mostly unknown. In terms of mineral and heavy metal content, the OBLL system accumulated most of these elements, except for Na, Ca, and K, which showed higher levels in the FLL (Table 6). The elevated concentrations of heavy metals in seaweed from off-bottom systems may be attributed to the absorption and accumulation of greater amounts of heavy metals from the sediment (Banik et al., 2023). Multiple studies have demonstrated that heavy metals tend to accumulate in higher amounts in bottom sediments compared to water columns due to their binding with organic materials and settling at the sediment layer (Karthick et al., 2012; Rashid et al., 2014; Raknuzzaman et al., 2018; Banik et al., 2023). This accumulation in seaweed can lead to increased toxicity risks at higher trophic levels of the food chain, posing potential health concerns for humans (Kumar and Sharma, 2021; Rajaram et al., 2020; Filippini et al., 2021).

5. Conclusion

The study aimed to evaluate the production performance and biochemical profile of *Ulva lactuca* under different cultivation systems and site-specific ecological conditions. Our findings indicate that the floating long-line (FLL) system demonstrated the highest production performance and superior nutritional quality, with elevated levels of protein and beneficial fatty acids. The off-bottom long-line (OBLL) system, while effective, showed lower nutrient content and higher levels of saturated fatty acids. Potential issues related to metal accumulation were also identified, highlighting the need for careful monitoring. These results suggest that the FLL system is the most efficient method for cultivating *U. lactuca*, offering significant benefits for sustainable farming practices. Future research should continue to explore the environmental and economic aspects of *U. lactuca* cultivation to enhance its viability as a valuable resource.

CRediT authorship contribution statement

Farjana Akter Chamily: Writing – original draft, Software, Methodology, Formal analysis, Data curation. **Mohammad Shakline Mostakim:** Methodology, Data curation. **Md Moshir Rahman:** Writing – review & editing, Software, Formal analysis, Data curation. **Sazeed Mehrab Souhardya:** Writing – review & editing, Methodology. **Md Sadequr Rahman Khan:** Writing – review & editing, Supervision. **Md Asaduzzaman:** Writing – review & editing, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare no competing or financial interests.

Data availability

Data will be made available on request.

Acknowledgements

This work was undertaken as a part of the One CGIAR Research Program Resilient Agri-Food System (RAFS). The field work was carried out under a sub-project of the United States Agency for International

Development (USAID)-funded Enhanced Coastal Fisheries in Bangladesh II (ECOFISH II) activity through a collaborative agreement between WorldFish Bangladesh and the Chattogram Veterinary and Animal Science University (CVASU). This study was also partially funded by the Ministry of Science and Technology, Bangladesh, under the allocation of special research grants for scientific research programs (SRG-231192), particularly for all of the costs associated with different biochemical analyses. Thanks to Dr. Mohammad Mokarrom Hossain, Chief of Party, ECOFISH-II project, for his support and encouragement. Sincere thanks are extended to the research associates and research assistants of the ECOFISH II project for their help in collecting various data and field research implementation for this study.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.aquaculture.2024.741436>.

References

- Abdullah, Al-M, Akhtar, A., Rahman, M.F., Kamal, A.H.M., Karim, N.U., Hassan, M.L., 2020. Habitat structure and diversity patterns of seaweeds in the coastal waters of Saint Martin's island, bay of Bengal, Bangladesh. *Reg. Stud. Mar. Sci.* 33, 100959.
- Abdulwahab, S., Rabee, A.M., 2015. Ecological factors affecting the distribution of the zooplankton community in the Tigris River at Baghdad region. *Egypt. J. Aquat. Res.* 41, 187–196.
- Abudabos, A.M., Okab, A.B., Aljumaah, R.S., Samara, E.M., Abdoun, K.A., Al-Haidary, A. A., 2013. Nutritional value of green seaweed (*Ulva lactuca*) for broiler chickens. *Ital. J. Anim. Sci.* 12, e28.
- Ajayan, K.V., Shamna, P., Haritha, D.S., Harilal, C.C., 2021. Fatty acid profile and metal composition of seaweeds from three locations of the continental shelf of Kerala coast, India. *Reg. Stud. Mar. Sci.* 45, 101864.
- Ale, M.T., Mikkelsen, J.D., Meyer, A.S., 2011. Differential growth response of *Ulva lactuca* to ammonium and nitrate assimilation. *J. Appl. Phycol.* 23, 345–351.
- Al-Hafedh, Y.S., Alam, A., Buschmann, A.H., 2015. Bioremediation potential, growth and biomass yield of the green seaweed, *Ulva lactuca* in an integrated marine aquaculture system at the Red Sea coast of Saudi Arabia at different stocking densities and effluent flow rates. *Rev. Aquac.* 7, 161–171.
- Ali, S.A., Anshary, H., Tahya, A.M., 2017. Environmental parameters and specific growth of *Kappaphycus alvarezii* in Saugi Island, South Sulawesi Province, Indonesia. *Aquacult. Aquarium Conserv. Legis.* 10, 698–702.
- Ansari, A.A., Ghanem, S.M., 2017. Seasonal variation in the growth responses of some chlorophytic algal flora of the Red Sea. *Egypt. J. Aquat. Res.* 43, 129–134.
- Ansari, A.A., Khan, F.A., 2009. Remediation of eutrophied water using *Spirodela polyrrhiza* L. Shleid in controlled environment. *Pan-Am. J. Aquat. Sci.* 4, 52–54.
- Ansari, A.A., Khan, F.A., Gill, S.S., Varshney, J., 2011. Aquatic plant diversity in eutrophic ecosystems. In: Ansari, A., Singh Gill, S., Lanza, G., Rast, W. (Eds.), *Eutrophication: Causes, Springer, Consequences and Control*, pp. 247–263.
- AOAC, 2000. Official Methods of Analysis of the Association of Official Analytical Chemists, 17th ed. Washington, D.C.
- AOAC, 2005. Official Methods of Analysis of the Association of Official Analytical Chemist. Benyamin Franklin Station, Washington D.C.
- APHA, 2005. Standard Methods for the Examination of Water and Waste Water.
- Asaduzzaman, M., Rahi Noor, A., Rahman, M.M., Akter, S., Hoque, N.F., Shakil, A., Wahab, M.A., 2019. Reproductive biology and ecology of the green mussel *Perna viridis*: a multidisciplinary approach. *Biology* 8, 88.
- Asaduzzaman, M., Akter, S., Hoque, N.F., Shakil, A., Noor, A.R., Akter, M.N., Rahman, M. M., 2020. Multifaceted linkages among eco-physiological factors, seasonal plankton dynamics and selective feeding behavior of the green mussel (*Perna viridis*) in the south-east coast of the bay of Bengal. *J. Sea Res.* 164, 101933.
- Banik, U., Mohiuddin, M., Wahab, M.A., Rahman, M.M., Nahiduzzaman, M., Sarker, S., Wong, L., Asaduzzaman, M., 2023. Comparative performances of different farming systems and associated influence of ecological factors on *Gracilaria* sp. seaweed at the south-east coast of the bay of Bengal, Bangladesh. *Aquaculture* 574, 739675.
- Baweja, P., Kumar, S., Sahoo, D., Levine, I., 2016. Biology of seaweeds. In: *Seaweed in Health and Disease Prevention*. Academic Press, pp. 41–106.
- Benjama, O., Masniyom, P., 2012. Biochemical composition and physicochemical properties of two red seaweeds (*Gracilaria fisheri* and *G. tenuistipitata*) from the Pattani Bay in southern Thailand. *Songklanakarin J. Sci. Technol.* 34, 223.
- Besada, V., Andrade, J.M., Schultze, F., González, J.J., 2009. Heavy metals in edible seaweeds commercialised for human consumption. *J. Mar. Syst.* 75, 305–313.
- Bhatia, S., Sharma, K., Dahiya, R., Bera, T., 2015. Modern Applications of Plant Biotechnology in Pharmaceutical Sciences. Academic press.
- Billah, M.M., Kader, M.A., Mahmud, S.S., Asif, A.A., Siddiqui, A.M., 2018. Diversity and distribution of seaweeds in Saint Martin Island, Bangladesh. *Int. J. Fish. Aquat. Stud.* 6, 166–169.
- Bolton, J.J., Robertson-Andersson, D.V., Shuuluka, D., Kandjengo, L., 2009. Growing *Ulva* (Chlorophyta) in integrated systems as a commercial crop for abalone feed in South Africa: a SWOT analysis. *J. Appl. Phycol.* 21, 575–583.
- Brien, R.O., Hayes, M., Sheldrake, G., Tiwari, B., Walsh, P., 2022. Macroalgal proteins: a review. *Foods* 11, 571.
- Britton, D., Schmid, M., Revill, A.T., Virtue, P., Nichols, P.D., Hurd, C.L., Mundy, C.N., 2021. Seasonal and site-specific variation in the nutritional quality of temperate seaweed assemblages: implications for grazing invertebrates and the commercial exploitation of seaweeds. *J. Appl. Phycol.* 33, 603–616.
- Bruhn, A., Dahl, J., Nielsen, H.B., Nikolaisen, L., Rasmussen, M.B., Markager, S., Olesen, B., Arias, C., Jensen, P.D., 2011. Bioenergy potential of *Ulva lactuca*: biomass yield, methane production and combustion. *Bioresour. Technol.* 102, 2595–2604.
- Bruhn, A., Tørring, D.B., Thomsen, M., Canal-Vergés, P., Nielsen, M.M., Rasmussen, M.B., Eybye, K.L., Larsen, M.M., Balsby, T.J.S., Petersen, J.K., 2016. Impact of environmental conditions on biomass yield, quality, and bio-mitigation capacity of *Saccharina latissima*. *Aquac. Environ. Interact.* 8, 619–636.
- Budidaya, D., 2005. Profil Rumpun Laut Indonesia. Jakarta, Direktorat Perikanan Budidaya, Departemen Kelautan dan Perikanan.
- Bunsom, C., Prathep, A., 2012. Effects of salinity, light intensity and sediment on growth, pigments, agar production and reproduction in *Gracilaria tenuistipitata* from Songkhla lagoon in Thailand. *Phycol. Res.* 60, 169–178.
- Buschmann, A.H., Hernandez-Gonzalez, M.D.C., Varela, D., 2008. Seaweed future cultivation in Chile: perspectives and challenges. *Int. J. Environ. Pollut.* 33, 432–456.
- Campbell, I., Macleod, A., Sahlmann, C., Neves, L., Funderud, J., Øverland, M., Hughes, A.D., Stanley, M., 2019. The environmental risks associated with the development of seaweed farming in Europe-prioritizing key knowledge gaps. *Front. Mar. Sci.* 6, 107.
- Cardozo, K.H., Vessecchi, R., Carvalho, V.M., Pinto, E., Gates, P.J., Colepicolo, P., Galembeck, S.E., Lopes, N.P., 2008. A theoretical and mass spectrometry study of the fragmentation of mycosporine-like amino acids. *Int. J. Mass Spectrom.* 273, 11–19.
- Carl, C., de Nys, R., Paul, N.A., 2014. The seeding and cultivation of a tropical species of filamentous *Ulva* for algal biomass production. *PLoS One* 9, e98700.
- Castelar, B., Reis, R.P., dos Santos Calheiros, A.C., 2014. *Ulva lactuca* and *U. flexuosa* (Chlorophyta, Ulvophyceae) cultivation in Brazilian tropical waters: recruitment, growth, and ulvan yield. *J. Appl. Phycol.* 26, 1989–1999.
- Cecchi, F., Pavan, P., Mata-Alvarez, J., 1996. Anaerobic co-digestion of sewage sludge: application to the macroalgae from the Venice lagoon. *Resour. Conserv. Recycl.* 17, 57–66.
- Chowdhury, M.S.N., Hossain, M.S., Aftabuddin, S., Alamgir, M., Sharifuzzaman, S.M., 2022. Seaweed aquaculture in Bangladesh: present status, challenges and future prospects. *Ocean Coast. Manag.* 228, 106309.
- Corino, C., Modina, S.C., Di Giancamillo, A., Chiapparini, S., Rossi, R., 2019. Seaweeds in pig nutrition. *Animals* 9, 1126.
- Dag, O., Dolgun, A., Konar, N.M., 2018. Onewaytests: an R package for one-way tests in independent groups designs. *R J.* 10.
- Dawczynski, C., Schubert, R., Jahreis, G., 2007. Amino acids, fatty acids, and dietary fibre in edible seaweed products. *Food Chem.* 103, 891–899.
- De Wrachien, D., Schultz, B., Goli, M.B., 2021. Impacts of population growth and climate change on food production and irrigation and drainage needs: a world-wide view. *Irrig. Drain.* 70, 981–995.
- Dhargalkar, V.K., Pereira, N., 2005. Seaweed: promising plant of the millennium. *Sci. Cult.* 71, 60–66.
- Dominguez, H., Loret, E.P., 2019. *Ulva lactuca*, a source of troubles and potential riches. *Mar. Drugs* 17, 357.
- Doty, M.S., 1971. Measurement of water movement in reference to benthic algal growth. *Bot. Mar.* 63, 309–318.
- Doty, M.S., 1986. Estimating returns from producing *Gracilaria* and *Eucheuma* on line farms. *Monogr. Biol.* 4, 45–62.
- Eggert, A., 2012. Seaweed responses to temperature. In: Wiencke, C., Bischof, K. (Eds.), *Seaweed Biology: Novel Insights into Ecophysiology, Ecology and Utilization*. Springer, pp. 47–66.
- El-Shafai, S.A., El-Gohary, F.A., Nasr, F.A., Van Der Steen, N.P., Gijzen, H.J., 2007. Nutrient recovery from domestic wastewater using a UASB-duckweed ponds system. *Bioresour. Technol.* 98, 798–807.
- Fannin, K.F., 1983. Marine Biomass Program: Anaerobic Digestion Systems Development and Stability Study.
- FAO, 2022. The State of World Fisheries and Aquaculture 2022. Towards Blue Transformation, Rome.
- Filippini, M., Baldisserotto, A., Menotta, S., Fedrizzi, G., Rubini, S., Gigliotti, D., Valpiani, G., Buzzi, R., Manfredini, S., Vertuani, S., 2021. Heavy metals and potential risks in edible seaweed on the market in Italy. *Chemosphere* 263, 127983.
- Foster, B.A., 2018. Barnacle ecology and adaptation. *Barnacle Biology*. Routledge 113–133.
- Fox, J., Weisberg, S., 2019. An R Companion to Applied Regression, 3rd ed. Sage Publication.
- Frøehlich, H.E., Afflerbach, J.C., Frazier, M., Halpern, B.S., 2019. Blue growth potential to mitigate climate change through seaweed offsetting. *Curr. Biol.* 29, 3087–3093.
- Fróna, D., Szenderák, J., Harangi-Rákos, M., 2019. The challenge of feeding the world. *Sustainability* 11, 5816.
- Ganesan, M., Sahu, N., Eswaran, K., 2011. Raft culture of *Gracilaria edulis* in open sea along the south-eastern coast of India. *Aquaculture* 321, 145–151.
- Gao, G., 2016. Developing Systems for the Commercial Culture of *Ulva* Species in the UK (Doctoral dissertation). Newcastle University.
- Ghaderiadekani, F., 2019. *Ulva* Growth, Development and Applications (Doctoral dissertation). University of Birmingham.
- Goecke, F., Labes, A., Wiese, J., Imhoff, J.F., 2010. Chemical interactions between marine macroalgae and bacteria. *Mar. Ecol. Prog. Ser.* 409, 267–299.

- Groenendijk, F.C., Bikker, P., Blaauw, R., Brandenburg, W.A., van den Burg, S.W.K., Harmsen, P.F.H., Jak, R.G., Kamermans, P., Van Krimpen, M.M., Prins, H., Wald, J., 2016. North-Sea-Weed-Chain: Sustainable Seaweed from the North Sea; an Exploration of the Value Chain (IMARES).
- Guarattini, T., Cardozo, K.H., Pinto, E., Colepicolo, P., 2009. Comparison of diode array and electrochemical detection in the C30 reverse phase HPLC analysis of algae carotenoids. *J. Braz. Chem. Soc.* 20, 1609–1616.
- Guebel, D.V., Nudel, B.C., Giulietti, A.M., 1991. A simple and rapid micro-Kjeldahl method for total nitrogen analysis. *Biotechnol. Tech.* 5, 427–430.
- Hagan, M., Anyangwe, N., 2023. Vitamin content in seaweeds: a systematic review on water-soluble and fat-soluble vitamins for adult daily intake. *Funct. Food Sci.* 3, 305–316.
- Harley, C.D., Anderson, K.M., Demes, K.W., Jorve, J.P., Kordas, R.L., Coyle, T.A., Graham, M.H., 2012. Effects of climate change on global seaweed communities. *J. Phycol.* 48, 1064–1078.
- Hoq, M.E., Haque, M.A., Islam, M.M., 2016. Feasibility of seaweed culture in Inani and Bakkhali coast of Cox's Bazar. *Bangladesh. Pak. J. Mar. Sci.* 25, 27–36.
- Hossain, M.S., Chowdhury, M.S.N., Alamgir, M., 2020. Seaweeds for blue economy in Bangladesh. *Food and Agriculture Organization of the United Nations*, 87pp.
- Hossain, M.S., Sharifuzzaman, S.M., Nobil, M.N., Chowdhury, M.S.N., Sarker, S., Alamgir, M., Uddin, S.A., Chowdhury, S.R., Rahman, M.M., Rahman, M.S., Chowdhury, S., 2021. Seaweeds farming for sustainable development goals and blue economy in Bangladesh. *Mar. Policy* 128, 104469.
- Hung, L.D., Hori, K., Nang, H.Q., Kha, T., Hoa, L.T., 2009. Seasonal changes in growth rate, carrageenan yield and lectin content in the red alga *Kappaphycus alvarezii* cultivated in Camranh Bay. *Vietnam. J. Appl. Phycol.* 21, 265–272.
- Hurd, C.L., 2000. Water motion, marine macroalgal physiology, and production. *J. Appl. Phycol.* 36, 453–472.
- Hurd, C.L., Harrison, P.J., Bischof, K., Lobban, C.S., 2014. *Seaweed Ecology and Physiology*. Cambridge University Press.
- Islam, A.K.M.N., 1998. The seaweed resources of Bangladesh. In: Ohno, M. (Ed.), *Critchley, A. T. Seaweed Resources of the World*, Japan International Cooperation Agency, pp. 5–25.
- Islam, A.N., Aziz, A., 1987. Addition to the list of marine algae of St. Martin's island. *Bangladesh. III: Red algae. Nova Hedwigia.* 45, 211–221.
- Islam, M.M., Shamsuddoha, M.D., 2018. Coastal and marine conservation strategy for Bangladesh in the context of achieving blue growth and sustainable development goals (SDGs). *Environ. Sci.* 87, 45–54.
- Islam, M.M., Khan, M.S.K., Hasan, J., Mallick, D., Hoq, M.E., 2017. Seaweed *Hypnea* sp. culture in Cox's Bazar coast. *Bangladesh. J. Zool.* 45, 37–46.
- Islam, M.M., Hoq, M.E., Haque, M.A., Khan, M.S.K., Hasan, J., Ali, M.Z., Mahmud, Y., 2019. Seaweeds of Bangladesh Coast. *Bangladesh Fisheries Research Institute, Mymensingh, Bangladesh*.
- Jansen, H.M., Bernard, M.S., Nederlof, M.A., van der Meer, I.M., van der Werf, A., 2022. Seasonal variation in productivity, chemical composition and nutrient uptake of *Ulva* spp. (Chlorophyta) strains. *J. Appl. Phycol.* 34, 1649–1660.
- Jin, S., Liu, Y., Sun, C., Wei, X., Li, H., Han, Z., 2018. A study of the environmental factors influencing the growth phases of *Ulva prolifera* in the southern Yellow Sea. *China. Mar. Pollut. Bull.* 135, 1016–1025.
- Kamlasi, Y., 2008. *Ecological and Biological Studies for the Development of Seaweed (Eucheuma Cottonii) in West Kupang District*. Kupang Regency, East Nusa Tenggara Province, Essay, IPB.
- Karthick, P., Sankar, R.S., Kaviarasana, T., Mohanraju, R., 2012. Ecological implications of trace metals in seaweeds: bio-indication potential for metal contamination in Wandoor, south Andaman Island. *Egypt. J. Aquat. Res.* 38, 227–231.
- Kirkendale, L., Saunders, G.W., Winberg, P., 2013. A molecular survey of *Ulva* (Chlorophyta) in temperate Australia reveals enhanced levels of cosmopolitanism. *J. Phycol.* 49, 69–81.
- Klyachko-Gurvich, G.L., Tsoqlin, L.N., Doucha, J., Kopetskii, J., Shebalina, I.B., Semenenko, V.E., 1999. Desaturation of fatty acids as an adaptive response to shifts in light intensity. *Physiol. Plant.* 107, 240–249.
- Kraan, W., 2018. *The Influence of Nutrient Availability, Salinity, Temperature and the Underwater Light Climate on the Growth of Seaweed (Doctoral Dissertation)*.
- Kumar, M.S., Sharma, S.A., 2021. Toxicological effects of marine seaweeds: a cautious insight for human consumption. *Crit. Rev. Food Sci. Nutr.* 61, 500–521.
- Kumar, M., Kumari, P., Trivedi, N., Shukla, M.K., Gupta, V., Reddy, C.R.K., Jha, B., 2011. Minerals, PUFAs and antioxidant properties of some tropical seaweeds from Saurashtra coast of India. *J. Appl. Phycol.* 23, 797–810.
- Lalegerie, F., Gager, L., Stiger-Pouvreau, V., Connan, S., 2020. The stressful life of red and brown seaweeds on the temperate intertidal zone: effect of abiotic and biotic parameters on the physiology of macroalgae and content variability of particular metabolites. In: *Advances in botanical research*. Academic Press, pp. 247–287.
- Lamare, M.D., Wing, S.R., 2001. Calorific content of New Zealand marine macrophytes. *N. Z. J. Mar. Freshw. Res.* 35, 335–341.
- Landis, J., 2024. Ggside: side grammar graphics. R package version 0 (3), 1.
- Lê, S., Josse, J., Husson, F., 2008. FactoMineR: a package for multivariate analysis. *J. Stat. Softw.* 25, 1–18.
- Lenth Russell, V., Paul, B., Maxime, H., Maarten, J., Jonathon, L., Fernando, M., Hannes, R., Henrik, S., 2022. Emmeans: estimated marginal means, aka least-squares means. R package version 1 (7), 5.
- Li, J.Y., Yang, F., Jin, L., Wang, Q., Yin, J., He, P., Chen, Y., 2018. Safety and quality of the green tide algal species *Ulva prolifera* for option of human consumption: a nutrition and contamination study. *Chemosphere* 210, 1021–1028.
- Li, X., Sun, X., Gao, L., Xu, J., Gao, G., 2021. Effects of periodical dehydration on biomass yield and biochemical composition of the edible red algae *Pyropia yezoensis* grown at different salinities. *Algal Res.* 56, 102315.
- Littler, M.M., 1980. Morphological form and photosynthetic performances of marine macroalgae: tests of a functional/form hypothesis. *Bot. Mar.* 22, 161–165.
- Lourenço, S.O., Barbarino, E., De-Paula, J.C., 2002. da S. Pereira, L.O.; Marquez, UML amino acid composition, protein content and calculation of nitrogen-to-protein conversion factors for 19 tropical seaweeds. *Phycol. Res.* 50, 233–241.
- Lu, Q., He, Z.L., Graetz, D.A., Stoffella, P.J., Yang, X., 2010. Phytoremediation to remove nutrients and improve eutrophic stormwaters using water lettuce (*Pistia stratiotes*). *Environ. Sci. Pollut. Res.* 17, 84–96.
- Luo, M.B., Liu, F., 2011. Salinity-induced oxidative stress and regulation of antioxidant defense system in the marine macroalga *Ulva prolifera*. *J. Exp. Mar. Biol. Ecol.* 409, 223–228.
- Manivannan, K., Devi, G.K., Thirumaran, G., Anantharaman, P., 2008. Mineral composition of marine macroalgae from Mandapam coastal regions; southeast coast of India. *Am. Eurasian J. Bot.* 1, 58–67.
- Mantri, V.A., Thakur, M.C., Kumar, M., Reddy, C.R.K., Jha, B., 2009. The carpospore culture of industrially important red alga *Gracilaria dura* (Gracilariaceae, Rhodophyta). *Aquaculture* 297, 85–90.
- Marinho-Soriano, E., Panucci, R.A., Carneiro, M.A.A., Pereira, D.C., 2009. Evaluation of *Gracilaria caudata* J. Agardh for bioremediation of nutrients from shrimp farming wastewater. *Bioresour. Technol.* 100, 6192–6198.
- Martínez, B., Pato, L.S., Rico, J.M., 2012. Nutrient uptake and growth responses of three intertidal macroalgae with perennial, opportunistic and summer-annual strategies. *Aquat. Bot.* 96, 14–22.
- Masyahoro, M., Mappiratu, M., 2010. Respon pertumbuhan pada berbagai kedalaman bibit dan umur panen rumput laut *Eucheuma cottonii* Di Perairan Teluk Palu. *Media Litbang Sulawesi Tengah.* 3, 151172.
- Mata, L., Magnusson, M., Paul, N.A., de Nys, R., 2016. The intensive land-based production of the green seaweeds *Derbesia tenuissima* and *Ulva ohnoi*: biomass and bioproducts. *J. Appl. Phycol.* 28, 365–375.
- Messyasz, B., Rybak, A., 2011. Abiotic factors affecting the development of *Ulva* sp. (Ulvothycaceae: Chlorophyta) in freshwater ecosystems. *Aquat. Ecol.* 45, 75–87.
- Mišurová, L., 2011. Chemical composition of seaweeds. *Handbook of Marine Macroalgae*. Wiley 171–192.
- Mohiuddin, M., Banik, U., Iqbal, M.Z., Chamily, F.A., Rahman, M.M., Nahiduzzaman, M., Wahab, M.A., Rahman, M.A., Asaduzzaman, M., 2023. Influence of cultivation systems and associated environmental factors on the growth performance of *Hypnea musciformis* seaweed at the south-east coast of the bay of Bengal. *Bangladesh. Aquac. Rep.* 32, 101718.
- Morais, T., Inácio, A., Coutinho, T., Ministro, M., Cotas, J., Pereira, L., Bahcevandziev, K., 2020. Seaweed potential in the animal feed: a review. *J. Mar. Sci. Eng.* 8, 559.
- Moreda-Piñeiro, A., Peña-Vázquez, E., Bermejo-Barrera, P., 2012. Significance of the presence of trace and ultratrace elements in seaweeds. *Handbook of marine macroalgae*. Wiley 116–170.
- Msuya, F.E., Neori, A., 2002. *Ulva reticulata* and *Gracilaria crassa*: macroalgae that can biofilter effluent from tidal fishponds in Tanzania. *West. Indian Ocean J. Mar. Sci.* 1, 117–126.
- Msuya, F.E., Neori, A., 2008. Effect of water aeration and nutrient load level on biomass yield, N uptake and protein content of the seaweed *Ulva lactuca* cultured in seawater tanks. *J. Appl. Phycol.* 20, 1021–1031.
- Nazni, P., Deepa, S., 2015. Evaluation of proximate and pigment composition of red seaweed in south coastal regions of Tamilnadu. *Int. J. Mar. Sci.* 4, 38–41.
- Neish, I.C., 2008. Good agronomy practices for *Kappaphycus* and *Eucheuma*: including an overview of basic biology. *SEAPlant. Net monograph* no. HB2F 1008. V3.
- Neori, A., Cohen, I., Gordin, H., 1991. *Ulva lactuca* biofilters for marine fishpond effluents. II. Growth rate, yield and C: N ratio. *Bot. Mar.* 34, 483–489.
- Neori, A., Shpigel, M., Ben-Ezra, D., 2000. A sustainable integrated system for culture of fish, seaweed and abalone. *Aquaculture* 186, 279–291.
- Nissen, S.H., Juul, L., Bruhn, A., Søndergaard, J., Dalsgaard, T.K., 2024. The biochemical composition and its relation to color of *Ulva* spp. upon harvest time. *J. Appl. Phycol.* 1–13.
- Noor, A.R., Shakil, A., Hoque, N.F., Rahman, M.M., Akter, S., Talukder, A., Ahmad-Al-Nahid, S., Wahab, M.A., Nahiduzzaman, M., Rahman, M.J., Asaduzzaman, M., 2021. Effect of eco-physiological factors on biometric traits of green mussel *Perna viridis* cultured in the south-east coast of the bay of Bengal. *Bangladesh. Aquacult. Rep.* 19, 100562.
- Nunes, N., Ferraz, S., Valente, S., Barreto, M.C., Pinheiro de Carvalho, M.A.A., 2017. Biochemical composition, nutritional value, and antioxidant properties of seven seaweed species from the Madeira archipelago. *J. Appl. Phycol.* 29, 2427–2437.
- Oliveira, E.C., Alveal, K., Anderson, R.J., 2000. Mariculture of the agar-producing Gracilarioid red algae. *Rev. Fish. Sci.* 8, 345–377.
- Ortiz, J., Romero, N., Robert, P., Araya, J., Lopez-Hernández, J., Bozzo, C., Navarrete, E., Osorio, A., Rios, A., 2006. Dietary fibre, amino acid, fatty acid and tocopherol contents of the edible seaweeds *Ulva lactuca* and *Durvillaea Antarctica*. *Food Chem.* 99, 98–104.
- Ovalle, A.R.C., Rezende, C.E., Carvalho, C.E.V., Jennerjahn, T.C., Ittekkot, V., 1999. Biogeochemical characteristics of coastal waters adjacent to small river–mangrove systems, East Brazil. *Geo-Mar. Lett.* 19, 179–185.
- Padhi, S., Swain, P.K., Behura, S.K., Baidya, S., Behera, S.K., Panigrahy, M.R., 2011. Cultivation of *Gracilaria verrucosa* (Huds) Papenfuss in Chilika Lake for livelihood generation in coastal areas of Orissa state. *J. Appl. Phycol.* 23, 151–155.
- Pappou, S., Dardavila, M.M., Savvidou, M.G., Louli, V., Magoulas, K., Voutsas, E., 2022. Extraction of bioactive compounds from *Ulva lactuca*. *J. Appl. Sci.* 12, 2117.
- Parker, H.S., 1981. Influence of relative water motion on the growth, ammonium uptake and carbon and nitrogen composition of *Ulva lactuca* (Chlorophyta). *Mar. Biol.* 63, 309–318.

- Pérez-Mayorga, D.M., Ladah, L.B., Zertuche-González, J.A., Leichter, J.J., Filonov, A.E., Lavín, M.F., 2011. Nitrogen uptake and growth by the opportunistic macroalga *Ulva lactuca* (Linnaeus) during the internal tide. *J. Exp. Mar. Biol. Ecol.* 406, 108–115.
- Peterson, B. G. Carl, P., 2020. PerformanceAnalytics: econometric tools for performance and risk analysis. R package version 2.0.4.
- Piazzini, L., Ceccherelli, G., Cinelli, F., 2001. Threat to macroalgal diversity: effects of the introduced green algae *Caulerpa racemosa* in the Mediterranean. *Mar. Ecol. Prog. Ser.* 210, 149–159.
- Prato, E., Parlapiano, I., Addis, P., Biondolino, F., 2017. Mediterranean seafood: Functional food properties. In: *The Mediterranean Diet: Perspectives. Food Components and Health Effects*. Nova Publisher, New York, pp. 171–187.
- Putra, B.D., Aryawati, R., 2011. Laju Pertumbuhan Rumput Laut *Gracilaria* sp. dengan Metode Penanaman yang Berbeda di Perairan Kalianda, Lampung Selatan. *Maspari J. Mar. Sci. Res.* 3, 36–41.
- R Development Core Team, 2023. R: A Language and Environment for Statistical Computing, Version 4.3.2. R Foundation for Statistical Computing, Vienna, Austria.
- Radiarta, I.N., Erlania, E., Rasidi, R., 2014. Analisis Pola Musim Tanam Rumput Laut, *Kappaphycus alvarezii* Melalui Pendekatan Kesesuaian Lahan Di Nusa Penida. *Bali. J. Ris. Akuakultur.* 9, 319–330.
- Rajaram, R., Rameshkumar, S., Anandkumar, A., 2020. Health risk assessment and potentiality of green seaweeds on bioaccumulation of trace elements along the Palk Bay coast. Southeastern India. *Mar. Pollut. Bull.* 154, 111069.
- Raknuzzaman, M., Al Mamun, M.H., Ahmed, M.K., Tokumura, M., Masunaga, S., 2018. Monitoring of seasonal variation of some trace metals concentration in surface water collected from the coastal area of Bangladesh. *J. Biodivers. Conserv. Bioresour. Manag.* 4, 67–80.
- Rashid, T., Hoque, S., Akter, S., 2014. Pollution in the bay of Bengal: impact on marine ecosystem. *Open J. Mar. Sci.* 5, 55.
- Rasyid, A., 2017. Evaluation of nutritional composition of the dried seaweed *Ulva lactuca* from Pameungpeuk waters. Indonesia. *Trop. Life Sci. Res.* 28, 119.
- Reddy, C.R.K., Jha, B., Fujita, Y., Ohno, M., 2008. Seaweed micropropagation techniques and their potentials: an overview. *J. Appl. Phycol.* 20, 609–617.
- Rezayian, M., Niknam, V., Ebrahimzadeh, H., 2019. Oxidative damage and antioxidative system in algae. *Toxicol. Rep.* 6, 1309–1313.
- Robertson-Andersson, D.V., Potgieter, M., Hansen, J., Bolton, J.J., Troell, M., Anderson, R.J., Halling, C., Probyn, T., 2008. Integrated Seaweed Cultivation on an Abalone Farm in South Africa. *J. Appl. Phycol.* p. 20.
- Robin, A., Sack, M., Israel, A., Frey, W., Müller, G., Golberg, A., 2018. Deashing macroalgae biomass by pulsed electric field treatment. *Bioresour. Technol.* 255, 131–139.
- Roleda, M.Y., Hurd, C.L., 2019. Seaweed nutrient physiology: application of concepts to aquaculture and bioremediation. *Phycologia* 58, 552–562.
- Ryther, J.H., Debusk, T.A., Blakeslee, M., 1984. Cultivation and Conversion of Marine Macroalgae (*Gracilaria* and *Ulva*). Harbor Branch Foundation, Fort Pierce, FL (USA).
- Sadigov, R., 2022. Rapid growth of the world population and its socioeconomic results. *Sci. World J.* 1, 8110229.
- Sari-Chmayssem, N., Taha, S., Mawlawi, H., Guégan, J.P., Jeftić, J., Benvegna, T., 2019. Extracted ulvans from green algae *Ulva linza* of Lebanese origin and amphiphilic derivatives: evaluation of their physico-chemical and rheological properties. *J. Appl. Phycol.* 31, 1931–1946.
- Sarkar, M.S.I., Kamal, M., Hasan, M.M., Hossain, M.I., 2019. Quality aspects of some value-added seaweed food and functional food products. *Fundam. Appl. Agric.* 4, 798–805.
- Sarker, S., Akter, M., Rahman, M.S., Islam, M.M., Hasan, O., Kabir, M.A., Rahman, M.M., 2021. Spatial prediction of seaweed habitat for mariculture in the coastal area of Bangladesh using a generalized additive model. *Algal Res.* 60, 102490.
- Sfriso, A., Pavoni, B., Marcomini, A., Orto, A.A., 1992. Macroalgae, nutrient cycles, and pollutants in the lagoon of Venice. *Estuaries* 15, 517–528.
- Sharma, R.C., Singh, N., Chauhan, A., 2016. The influence of physio-chemical parameters on phytoplankton distribution in a head water stream of Garhwal Himalayas: a case study. *Egypt. J. Aquat. Res.* 42, 11–21.
- Siddiqui, A.A.M., Kashem, M.A., Mondal, M.A.I., Shafiuddin, M., 2019. Commercially important seaweed cultivation and its potentials for the coastal areas of Cox's Bazar, Bangladesh. *Int. J. Fish. Aquat. Stud.* 7, 463–470.
- Sobuj, M.K.A., Mostofa, M.G., Islam, Z., Rabby, A.F., Rahman, T., Sonia, S.S., Hasan, S.J., Rahman, S., 2023. Floating raft culture of *Gracilariopsis longissima* for optimum biomass yield performance on the coast of Cox's Bazar. Bangladesh. *Sci. Rep.* 13, 2308.
- Stengel, D.B., Connan, S., Popper, Z.A., 2011. Algal chemodiversity and bioactivity: sources of natural variability and implications for commercial application. *Biotechnol. Adv.* 29, 483–501.
- Tasnim, R., Sarker, S., Chamily, F.A., Mohiuddin, M., Ferdous, A., Haque, A.M., Nahiduzzaman, M., Wahab, M.A., Rahman, M.M., Asaduzzaman, M., 2024. Site suitability mapping for different seaweed cultivation systems along the coastal and marine waters of Bangladesh: a generalized additive modelling approach for prediction. *Algal Res.* 74, 103404.
- Taylor, R., Fletcher, R.L., Raven, J.A., 2001. Preliminary studies on the growth of selected 'green tide' algae in laboratory culture: effects of irradiance, temperature, salinity and nutrients on growth rate. *Bot. Mar.* 44, 327–336.
- Ullah, M.R., Islam, M.A., Khan, A.B.S., Bosu, A., Yasmin, F., Hasan, M.M., Islam, M.M., Rahman, M.A., Mahmud, Y., 2023. Effect of stocking density and water depth on the growth and production of red seaweed, *Gracilaria tenuistipitata* in the Kuakata coast of Bangladesh. *Aquacult. Rep.* 29, 101509.
- Vinagre, P.A., Simas, T., Cruz, E., Pinori, E., Svenson, J., 2020. Marine biofouling: a European database for the marine renewable energy sector. *J. Mar. Sci. Eng.* 8, 495.
- Wahl, M., Goecke, F., Labes, A., Dobretsov, S., Weinberger, F., 2012. The second skin: ecological role of epibiotic biofilms on marine organisms. *Front. Microbiol.* 3, 292.
- Wang, W.J., Sun, X.T., Liu, F.L., Liang, Z.R., Zhang, J.H., Wang, F.J., 2016. Effect of abiotic stress on the gametophyte of *Pyropia katadae* var. *hemiphylla* (Bangiales, Rhodophyta). *J. Appl. Phycol.* 28, 469–479.
- Wickham, H., 2016. ggplot2: Elegant Graphics for Data Analysis. Springer International Publishing, New York.
- Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L.D.A., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., 2019. Welcome to the Tidyverse. *J. Open Source Softw.* 4, 1686.
- Wisnu Ariyati, R., Lakshmi Widowati, L., Rejeki, S., 2016. Performa produksi rumput laut *Euchema cottonii* yang dibudidayakan menggunakan metode long-line vertikal dan horizontal.
- Wu, H., Shin, S.K., Jang, S., Yarish, C., Kim, J., 2018. Growth and nutrient bioextraction of *Gracilaria chorda*, *G. vermiculophylla*, *Ulva prolifera*, and *U. compressa* under hypo- and hyper-osmotic conditions. *Algal Res.* 33, 329–340.
- WWF, 2014. Better Management Practices: Budidaya Rumput Laut *Gracilaria* Sp. WWF Press, Jakarta, di Tambak.
- Xia, J., Li, Y., Zou, D., 2004. Effects of salinity stress on PSII in *Ulva lactuca* as probed by chlorophyll fluorescence measurements. *Aquat. Bot.* 80, 129–137.
- Yaich, H., Garna, H., Besbes, S., Paquot, M., Blecker, C., Attia, H., 2011. Chemical composition and functional properties of *Ulva lactuca* seaweed collected in Tunisia. *Food Chem.* 128, 895–901.
- Yang, Y.F., Fei, X.G., Song, J.M., Hu, H.Y., Wang, G.C., Chung, I.K., 2006. Growth of *Gracilaria* sp. under different cultivation conditions and its effects on nutrient removal in Chinese coastal waters. *Aquaculture* 254, 248–255.
- Yıldız, G., Tiryaki, Ş., 2017. Physiological effects of light intensity on the opportunistic algae *Ulva rigida* C. Agardh (Chlorophyta). *Celal Bayar Univ. J. Sci.* 13, 631–634.
- Yuniarsih, E., Nirmala, K., Radiarta, I.N., 2015. The level of nitrogen and phosphorus absorption in IMTA (integrated multi-trophic aquaculture) based seaweed cultivation in Gerupuk Bay, Central Lombok, West Nusa Tenggara. *J. Ris. Akuakulture.* 9, 487–500.
- Zafar, M., 2004. Feasibility Study of Seaweed Culture in the Coastal Area of Bangladesh. Project Report, Ministry of Science and Information and Communication Technology, Government of Bangladesh.
- Zafar, M., 2005. Seaweed culture in Bangladesh holds promise. *INFOFISH Int.* 1, 8–10.
- Zafar, M., 2007. Seaweed culture (*Hypnea* sp.) in Bangladesh-seaweed culture, in: U.O.C. Institute of Marine Science and Fisheries (Ed.), Chittagong, p. 14.
- Zollmann, M., Liberzon, A., Palatnik, R.R., Zilberman, D., Golberg, A., 2023. Effects of season, depth and pre-cultivation fertilizing on *Ulva* growth dynamics offshore the eastern Mediterranean Sea. *Sci. Rep.* 13, 14784.